Highlights

CLEAR-ASD: Calibrated and Interpretable Autism Classification from Functional Connectomes

Reshad Ul Karim, Syeda Maliha Tabassum, Mehreen Mallick Fiona, Tareq Sujat, Rakin Bin Rabbani, Aniqua Nusrat Zereen

- One ABIDE-I pipeline fits every step inside the training split; AUROC is 0.776.
- Detection weakens as IQ rises, from AUROC 0.873 below 85 to 0.719 at 115 or above.
- Somatomotor hypoconnectivity strengthens when head motion is matched within site.
- Selecting connections before the split raises AUROC from 0.756 to 0.987.
- Linear classifiers reach 0.735 to 0.757, against 0.611 to 0.624 for tree ensembles.

CLEAR-ASD: Calibrated and Interpretable Autism Classification from Functional Connectomes

Reshad Ul Karim[a,*], Syeda Maliha Tabassum[a], Mehreen Mallick Fiona[a], Tareq Sujat[a], Rakin Bin Rabbani[a] and Aniqua Nusrat Zereen[b]

[a] *Department of Computer Science and Engineering, BRAC University, Kha 224 Bir Uttam Rafiqul Islam Avenue, Dhaka, 1212, Bangladesh*

[b] *Mahidol University, Thailand*

ARTICLE INFO

Keywords:
Autism spectrum disorder
Functional connectivity
Resting-state fMRI
Tangent space
Cross-validation
Data leakage

ABSTRACT

Reported performance in autism classification from resting-state functional magnetic resonance imaging (rs-fMRI) is difficult to compare across studies, because sample composition and analysis choices differ. What such performance measures is therefore examined on the Autism Brain Imaging Data Exchange I (ABIDE-I), using a single pipeline in which every fitted quantity is estimated within the training partition. Connectivity is described in three ways for each of 871 participants from 20 site labels, namely shrinkage correlation and static and windowed tangent-space connectivity, and three regularised logistic models are averaged. Discrimination reaches a participant-pooled area under the receiver operating characteristic curve (AUROC) of 0.776, and 0.774 when a whole acquisition site is withheld from training. In the wider quality-control cohort of 1,032 participants, an accuracy of 0.706 is obtained, matching published reports of 0.700 to 0.708. Six findings are established. Detection is weakest in cognitively able autistic individuals, because AUROC falls from 0.873 below full-scale intelligence quotient (IQ) 85 to 0.719 at 115 or above. Reduced somatomotor connectivity strengthens, rather than disappears, when participants are matched for head motion within site, and all 21 regions strengthen. Selecting 2,000 of 19,900 connections before the data are split raises AUROC from 0.756 to 0.987, whereas selecting whole regions does not. Linear classifiers reach 0.735 to 0.757, against 0.611 to 0.624 for tree ensembles. Classifier scores reflect group membership rather than the degree of autism, and the advantage of windowed connectivity is explained by covariance shrinkage. Reported accuracy therefore depends on sample composition and analysis procedure.

1. Introduction

Autism spectrum disorder (ASD) is recognized as a neurodevelopmental condition in which social communication and behavioral patterns are affected[why is the citation 39 at first?][39]. ASD diagnosis is dependent on behavioural assessment by trained clinicians, supported by the Autism Diagnostic Observation Schedule [36] and the Autism Diagnostic Interview–Revised [37]. Each assessment is conducted in person by trained clinicians through behavioural observation and clinical judgement, making the diagnostic process both time- and resource-intensive. Timely access to assessment is therefore important, as ASD can be identified earlier and appropriate interventions and support can be provided during critical stages of development, in line with current clinical recommendations. For this reason, brain-based assessment can be used as a complementary approach to address some of these practical limitations and support clinical diagnosis.

Resting-state functional magnetic resonance imaging (rs-fMRI) is used to measure spontaneous brain activity at rest. During the scan, spontaneous changes in blood-oxygen-level-dependent signals are recorded from different brain regions. These signals are then used to estimate functional connectivity, which is defined

*Corresponding author

reshad.sazid@gmail.com (R.U. Karim); syedamalihatabassum19@gmail.com (S.M. Tabassum); mehreen.mallick.fiona@g.bracu.ac.bd (M.M. Fiona); tareq.sujat@g.bracu.ac.bd (T. Sujat); rakin.rabbani@bracu.ac.bd (R.B. Rabbani); aniqua.zer@mahidol.ac.th (A.N. Zereen)

ORCID(s):

as the statistical relationship between the activity of different brain regions. In autistic individuals, differences in functional connectivity are observed across several brain regions and networks in large multisite studies. Functional connectivity can therefore be used as an input for group comparisons and ASD classification. For the development and evaluation of such classifiers, the Autism Brain Imaging Data Exchange I (ABIDE-I) is commonly used as a shared dataset. In ABIDE-I, rs-fMRI and phenotypic data from 1,112 participants across multiple sites are made openly available [19]. The dataset is based on a research population, in which participants are recruited for research studies and most individuals in the non-autistic group are typically developing controls.

The problem addressed in this study is that reported findings from ABIDE-I, including classification performance and group differences, can be influenced by sample composition and analysis choices that are not consistently controlled across studies. In systematic reviews of studies using ABIDE-I data, differences are reported in sample selection, preprocessing, validation methods, and classifier design, making direct comparison across studies difficult [26, 49]. Four such factors are examined in this study, with the first two related to characteristics of the study sample. One of these characteristics is cognitive ability. In published ABIDE-I benchmarks, classification performance is generally reported for the overall cohort, while differences in performance across levels of cognitive ability are rarely examined [3, 20, 24]. Full-scale intelligence quotient (IQ) varies across autistic participants and is generally lower than that of non-autistic participants in the dataset (Table 1). Classification performance may therefore be influenced by the cognitive characteristics of the participants included in the analysis. This possibility is further supported by reports that classifier scores vary with verbal IQ in ABIDE data [43]. This finding alone is not sufficient to establish whether the relationship is specifically associated with autism. A second sample-related factor is head motion, which is known to affect functional-connectivity measurements [46]. In this study, head motion is measured using mean framewise displacement, which represents the average movement of the head between consecutive image volumes. Greater head motion is observed among autistic participants in the dataset (Table 1), indicating that some reported differences in functional connectivity may also be influenced by motion. This issue is relevant to reported reductions in connectivity within the somatomotor system, which includes regions involved in movement and bodily sensation [25, 29]. Since only the linear effect of motion is removed through motion regression, it is not fully established whether this connectivity difference remains when autistic and non-autistic participants are explicitly matched for head motion.

The remaining two factors are related to choices made during the analysis. The first factor is the scope of feature selection, through which the features most relevant to ASD classification are identified. During cross-validation, one fold is held out for testing, while the remaining folds are used for model training. If features are selected using diagnostic labels from the full dataset before cross-validation, information from the test data is included in the training process. This results in data leakage and may lead to an overestimation of classification performance. The second factor is the choice of classifier. More complex nonlinear models have increasingly been proposed for ASD classification [35, 50, 51]. Strong performance with tangent-space connectivity and regularised linear classifiers has also been reported in an earlier benchmark [14]. In tangent-space connectivity, each participant's covariance matrix, which describes how signals from pairs of brain regions vary together, is represented relative to a reference matrix. The independent effects of feature-selection scope and classifier choice are difficult to determine from comparisons across existing studies because other parts of the analysis pipeline also differ. Therefore, their effects need to be evaluated using the same participants, features, and cross-validation folds.

Existing approaches to ASD classification using ABIDE-I are mainly focused on improving functional-connectivity features, classification models, and the handling of differences between acquisition sites. Different representations of functional connectivity, including the tangent-space projection of covariance matrices [54], are compared in a controlled benchmark [14]. Accuracies close to 0.70 are reported for deep autoencoder models on ABIDE-I [3, 20, 24]. Graph-based and convolutional models are also used to represent relationships between participants [44] and connections within brain networks [32]. These modelling approaches are further extended through multi-atlas deep ensembles [35], dynamic graph models [50], and multimodal graph models [51]. Differences between acquisition sites are addressed through harmonisation methods [21, 31] and validation on held-out sites [1]. Performance in most of these studies is reported for the whole cohort, so variation in classification performance across participant groups is not clearly established. It also remains unclear whether reported differences in functional connectivity are preserved when head

motion is directly controlled. In addition, the effects of feature-selection scope and classifier family are rarely examined under the same cross-validation setting, apart from the controlled benchmark reported by Dadi et al. [14]. These limitations support the need for a controlled evaluation in which participant characteristics and analysis choices are examined within the same framework. A detailed review of these existing approaches and their limitations is provided in Section 2.

To address this gap, a leakage-controlled evaluation is proposed, built on one fixed prediction pipeline (Figure 1). In this pipeline, three connectivity descriptions are computed for each participant: correlation, static tangent-space, and windowed tangent-space connectivity. The windowed description is computed from shorter segments, or windows, of each scan. Each description is then classified by ℓ_2-regularised logistic regression, a linear model whose weights are shrunk towards zero, and the three probabilities are averaged. Every learned quantity, including the tangent reference, the feature scaler, and the penalty strength, is fitted within the training partition. Performance is estimated under two schemes. In pooled ten-fold validation, participants from all sites are divided into ten folds; in leave-one-site-out validation, each of the 20 recorded site labels is held out once. Discrimination is summarised by the area under the receiver operating characteristic curve (AUROC). This is the probability that a randomly chosen autistic participant is scored above a randomly chosen non-autistic participant. Participant-pooled values are computed once from all held-out predictions, whereas fold-mean values are averaged across folds. Alongside this pipeline, six targeted analyses are run: cognitive-ability stratification against a fixed comparison group, group contrasts before and after within-site matching for head motion, feature-selection scope, classifier family, the scope of the symptom-severity associations, and a shrinkage control for the windowed description. The group contrasts are univariate and do not use the classifier. They are exploratory, like the cognitive-ability analysis, and are computed without global signal regression, which would remove the whole-brain mean signal (Sections 3.1.2 and 3.7). The selection-scope comparison, the symptom-association comparison, and the shrinkage control are, in contrast, primary descriptive comparisons, whereas the classifier comparison is a secondary one. These designations set the evidential weight of each result (Section 3.8.1).

The main contributions of this study are as follows.

- **Cognitive ability limits connectivity-based detection of autism.** Detection is shown to be weakest in cognitively able autistic individuals. Against a fixed comparison group, AUROC falls from 0.873 below full-scale IQ 85 to 0.719 at 115 or above. Sample composition therefore matters.

- **The somatomotor connectivity difference in autism is robust to head motion.** Reduced somatomotor connectivity strengthens, rather than disappears, when participants are matched for head motion within site. The mean regional effect strengthens from -0.215 to -0.249, supporting earlier reports under motion control.

- **Connection-level feature selection before the split inflates reported performance.** Selecting 2,000 of 19,900 connections with all labels raises AUROC from 0.756 to 0.987, whereas selecting whole regions has little effect. Without this step, accuracy reaches 0.706, matching published benchmarks (0.700–0.708).

- **Linear classifiers suit high-dimensional connectivity features.** On identical features and folds, linear classifiers reach AUROCs of 0.735–0.757, against 0.611–0.624 for fixed-configuration tree ensembles. Only the logistic model is tuned, so the gap between the families is not produced by tuning alone.

- **Classifier scores are shown to reflect group membership rather than the degree of autism.** Symptom-severity associations are obtained across both groups, whereas none survives correction within the autistic group. For the Social Responsiveness Scale total, the correlation falls from 0.438 to 0.050.

- **The advantage of windowed connectivity is explained by covariance shrinkage.** An additional shrinkage is applied by the windowed description alone. Once the whole-scan description receives it, that description gains 0.013 AUROC. The windowed advantage then disappears, leaving a difference of -0.002.

The remainder of the paper is organised as follows: related work is reviewed in Section 2, the methods are described in Section 3, and the results are reported in Section 4. The findings are then discussed in Section 5, and conclusions are drawn in Section 6.

2. Related Work

Research on autism spectrum disorder (ASD) classification using resting-state functional magnetic resonance imaging (rs-fMRI) has gradually developed from statistical modelling and functional-connectivity representation. More recent research is focused on multisite learning, deep architectures, dynamic connectivity, multimodal fusion, and reliability-aware prediction. In early studies, methods are established for representing high-dimensional covariance and brain-network information. These methods are later extended through deep learning, multiple brain atlases, temporal modelling, and graph-based approaches to improve classification performance. Alongside these developments, calibration, interpretability, selective prediction, and statistical validation are increasingly used to improve the reliability of machine-learning predictions. Overall, the performance and reliability of ASD classification are shaped by several methodological choices, including how functional connectivity is represented, how variation across acquisition sites is controlled, and how prediction uncertainty is assessed.

2.1. Foundations of Functional Connectome-Based Analysis

The methodological basis for later connectome-based ASD classification is supported by several important statistical developments. Exact confidence limits for binomial outcomes are proposed in an early statistical study [10], providing a basis for the reliable estimation of classification error. Alongside the measurement of uncertainty, stable estimation of high-dimensional connectivity is also required. For this purpose, covariance estimation is improved through the Ledoit–Wolf shrinkage method [33]. This method is particularly relevant to rs-fMRI, where covariance matrices are estimated from a limited number of time points across many brain regions. Reliable comparison between classification models is further supported by statistical procedures developed for classifier evaluation [16]. As multisite datasets are increasingly used, variation between acquisition sites also needs to be controlled. The ComBat empirical-Bayes framework is originally proposed for the correction of batch effects [31]. About a decade later, the same framework is adapted to multi-site neuroimaging to reduce scanner- and site-related variation [21].

Building on these statistical foundations, the geometric structure of covariance matrices is considered more directly in later functional-connectivity research. Group-level covariance modelling is developed by treating covariance matrices according to their non-Euclidean geometry [54]. From this formulation, tangent-space representations are established for the analysis of functional connectivity and later become widely used in connectome-based prediction. A biological framework for interpreting these connectivity patterns is also provided through the identification of seven large-scale intrinsic brain networks [57]. Through this organisation, large numbers of individual connections can be interpreted within broader functional systems.

These developments provide a strong basis for static functional-connectivity analysis, where connectivity is estimated across the full duration of a scan. The temporal organisation of these connections is then examined through dynamic functional-connectivity research, in which changes in network relationships over time are shown to provide additional information [28]. Functional connectivity is therefore increasingly represented through both spatial patterns across brain regions and temporal changes in these patterns.

2.2. Multisite ASD Data and Reproducibility

Large-scale ASD neuroimaging research is substantially advanced through the release of the Autism Brain Imaging Data Exchange (ABIDE) [19]. Data from multiple international sites are combined, making larger ASD classification studies possible. This increased scale also introduces variation in scanner type, acquisition protocol, demographic composition, and site characteristics. As a result, prediction models may capture acquisition-related differences instead of ASD-related neural patterns.

The use of larger multisite datasets also requires high-dimensional brain information to be represented carefully. Regularised matrix regression is proposed for this purpose, where predictors are kept in matrix form so that relationships among features are preserved during modelling [58]. Careful analysis is also required when changes in connectivity over time are examined. Sliding-window estimates are shown to depend on

window length, and apparent connectivity changes may be produced by analysis choices [34]. Temporal information therefore needs to be evaluated with careful control of scan length and window design.

These methodological choices directly affect whether findings can be reproduced across datasets and sites. Using multisite resting-state data, biomarker performance is shown to depend on preprocessing, feature construction, and validation design [1]. The reliability of time-varying connectivity is also examined through a review of dynamic functional-connectivity methods [47]. Together, these studies show that stable ASD-related findings depend on careful control of participants, analysis choices, and acquisition sites.

2.3. Reliability, Calibration, and Explainability

Alongside neuroimaging research on representation and generalisation, reliability-oriented machine-learning methods are also developed. Probability calibration is examined in a study [23], where high classification accuracy is shown to be insufficient for reliable confidence estimates. A model may therefore rank participants correctly while producing probabilities that do not match actual prediction frequencies. Selective classification is introduced in a study [22], where uncertain cases are allowed to be rejected rather than requiring a prediction for every input. A trade-off between coverage and prediction error is therefore created, which is particularly relevant to screening applications where uncertain cases can be referred for further clinical assessment. Interpretability is further supported through SHAP [38] to measure the contribution of individual features to each prediction. Evaluation is therefore extended beyond conventional accuracy, as calibration is used to assess prediction confidence, selective prediction is used to determine when a prediction should be issued, and explanation methods are used to identify the contributing features. Clinical usefulness is also examined through decision-curve analysis [53], where statistical performance and practical decision value are evaluated separately. Reliable ASD classification is therefore increasingly evaluated through prediction accuracy, confidence, selective decision-making, interpretability, and clinical usefulness.

2.4. Deep Learning for ASD Classification

Alongside these reliability-focused developments, deep learning is increasingly applied to ABIDE functional-connectivity data. In a study [24], distributed connectivity patterns are learned automatically and an accuracy of approximately 70.0% is reported, showing that ASD-related information can be captured through deep feature learning. The effect of multisite variation still needs to be addressed when such models are applied to ABIDE. For this purpose, ComBat is adapted for neuroimaging harmonisation in a study [21], allowing scanner- and site-related variation to be reduced while biological variation is preserved. The harmonisation step also needs to be estimated without information from held-out test data. Deep representation learning is further developed through ASD-DiagNet, which is proposed in a study [20]. Representation learning and classification are combined, and an accuracy of approximately 70.3% is reported. The small difference in performance indicates that classification may also be influenced by the way functional connectivity is represented.

This role of connectivity representation is examined more directly in a controlled benchmarking study [14]. Multiple connectome representations and predictive models are compared under the same setting, and tangent-space connectivity is identified as a strong representation. With an appropriate connectivity representation, comparatively simple classifiers are shown to remain competitive with more complex nonlinear models. Sparse representation learning is further explored through ASD-SAENet, which is proposed in a study [3]. A sparse autoencoder is combined with a deep neural network, and an accuracy of approximately 70.8% is reported. Across these deep-learning approaches, the increase from 70.0% to 70.3% and then 70.8% remains relatively small. This pattern suggests that deeper representation learning alone may not fully address the challenges of heterogeneous multisite rs-fMRI data.

2.5. Evidence from Systematic Reviews

The broader performance of rs-fMRI machine-learning methods is examined through a systematic review and meta-analysis in [49]. Moderate pooled sensitivity, specificity, and discrimination are reported, while considerable methodological heterogeneity is identified. Differences in preprocessing, sample selection, validation strategy, feature extraction, and classifier design make direct comparison between studies difficult.

A later systematic review of deep-learning and rs-fMRI biomarkers [26] reports continued methodological development but identifies similar limitations related to generalisability, reproducibility, and inconsistent

evaluation. These reviews provide an important analytical perspective: higher accuracy reported by an individual model cannot be interpreted independently of the dataset definition and experimental pipeline used to obtain it.

Therefore, later ASD classification studies increasingly attempt to improve the representation itself rather than only increasing classifier depth.

2.6. Multi-Atlas and Dynamic Connectivity Models

Dependence on a single brain parcellation is addressed through MADE-for-ASD [35]. A multi-atlas deep ensemble is introduced, and approximately 75.2% accuracy is reported. Different atlases describe the same brain according to different regional boundaries, so their predictions may contain complementary information. Multi-atlas fusion therefore provides a possible way to reduce dependence on one predefined parcellation.

However, the improvement is introduced together with a more complex ensemble architecture. It therefore becomes difficult to determine whether performance is improved because of atlas diversity, deep-model capacity, demographic information, or their interaction. This motivates controlled evaluations in which atlas fusion is varied while the remaining pipeline is kept unchanged.

Dynamic functional connectivity is later revisited through a GNN-LSTM framework [50]. Graph neural networks are combined with temporal sequence modelling, and approximately 80.4% accuracy is reported on ABIDE-I. Compared with static approaches, connectivity variation across the scan is explicitly represented rather than being averaged into one matrix.

This development addresses an important limitation of static functional connectivity, but it introduces several additional decisions. Window length, overlap, temporal ordering, graph construction, and sequence-model complexity can all affect performance. Therefore, the contribution of temporal variation should be separated from the contribution of the classifier through controlled ablation.

2.7. Explainable and Multimodal ASD Models

Interpretability receives greater attention in recent ASD classification research. Explainable artificial intelligence is applied to ABIDE fMRI data in [55], where highly predictive performance is reported under the stated filtering condition and influential brain regions are identified through multiple explanation methods. Such work demonstrates the potential of connecting prediction with biologically interpretable patterns.

However, unusually high performance on heterogeneous multisite data also increases the importance of strict validation. Feature selection, filtering, preprocessing, and cross-validation must remain independent of held-out data. The broader lesson from earlier benchmarking [14] and meta-analysis [49] therefore remains relevant: reported performance must be interpreted together with the complete experimental procedure.

The progression toward richer representations continues through CAS-GNN [51]. Structural and functional MRI are integrated using cross-attention and subject-adaptive graph construction, and approximately 79.3% accuracy is reported. Compared with earlier functional-connectivity-only approaches, multimodal information is used to represent each participant more comprehensively.

Although multimodal graph learning increases representational capacity, additional learned components also make generalisation more difficult to assess. Performance gains must therefore be evaluated against cross-site stability, model complexity, and potential overfitting.

2.8. Research Gap

The literature shows two connected methodological developments. The first progresses from stable covariance estimation [33], geometric connectivity modelling [54], and network-level representation [57] toward dynamic connectivity [28, 34, 47], multisite analysis [19], reproducibility [1], deep learning [3, 20, 24], multi-atlas fusion [35], dynamic graph modelling [50], and multimodal graph learning [51]. Each stage addresses a limitation of the previous representation, but greater representational complexity also introduces additional sources of variation and overfitting.

The second development concerns prediction reliability. Statistical comparison and confidence estimation are formalised in [10, 16], while harmonisation [21, 31], calibration [23], selective prediction [22], explainability [38], and decision-oriented evaluation [53] show that classification accuracy alone is insufficient for a

screening-oriented system. Reliable probabilities, uncertainty-aware decisions, interpretable features, and performance on unseen acquisition sites are also required.

Despite these advances, these directions are commonly investigated separately. Dynamic connectivity, multi-atlas information, harmonisation, interpretability, calibration, selective prediction, and cross-site generalisation are rarely evaluated within the same controlled pipeline. It also remains unclear whether increasingly complex classifiers provide greater benefit than carefully designed functional-connectivity representations when identical validation conditions are used.

The present work is therefore designed around a controlled and reliability-oriented evaluation. Complementary functional-connectivity representations are compared under the same data partitions, tangent-space quantities are estimated from training data only, and cross-site performance is evaluated through leave-one-site-out validation. Structured low-rank connectivity patterns are used for interpretation, while calibrated probabilities and selective prediction are used to manage uncertainty. Each component is evaluated independently so that improvements can be attributed to the corresponding methodological choice rather than to model complexity alone.

3. Methods

The methodology follows a single pipeline from participant selection to statistical analysis. A primary cohort of 871 participants is first selected from ABIDE-I to reduce the influence of poor-quality scans and provide a more consistent sample for the primary analyses. Selection is based on the available anatomical and functional image-quality ratings from three raters, with participants retained only when the required ratings indicate acceptable, or where permitted borderline, image quality. Preprocessed regional time series are then obtained for each participant, representing changes in the signal of each brain region over the course of the scan.

Three functional-connectivity representations are constructed from these time series. The first captures whole-scan correlations between pairs of brain regions using shrinkage-based covariance estimation. The second represents whole-scan connectivity in tangent space by describing each participant's covariance pattern relative to a reference covariance estimated from the training data. The third extends this approach to shorter windows of the scan, allowing connectivity to be represented across multiple temporal segments. Each representation is standardised and classified separately using ℓ_2-regularised logistic regression, after which the three predicted probabilities are averaged to obtain the final ensemble probability.

Model performance is evaluated using both pooled ten-fold cross-validation and leave-one-site-out validation, while classifier regularisation is selected through nested cross-validation. All quantities required for model fitting are estimated from the corresponding training data to prevent information from the held-out data from entering the primary pipeline. The resulting out-of-fold predictions are subsequently used to assess discrimination, calibration, and selective prediction. Additional analyses examine the effects of alternative connectivity representations, classifiers, atlases, feature-selection strategies, cohort definitions, and participant subgroups, together with associations between classifier scores and symptom measures. Separate group-contrast analyses evaluate diagnostic-group differences in functional connectivity after adjustment for relevant nuisance variables. Fold-level results and out-of-fold predictions are retained to support paired statistical comparisons across experimental conditions. The complete methodological workflow is summarised in Figure 1.

Figure 1 summarises the primary modelling pipeline. A primary cohort of 871 participants is obtained from ABIDE-I after image-quality screening based on the available ratings from three raters. Preprocessed regional time series are then extracted from the C-PAC `filt_noglobal` derivative using the 200-region CC200 atlas. From these signals, three complementary functional-connectivity representations are constructed: shrinkage-based whole-scan correlation, static tangent-space connectivity, and windowed tangent-space connectivity computed from six temporal windows of 50 volumes. For the tangent-space representations, reference covariance matrices are estimated exclusively from the corresponding training data to prevent information from the held-out partition from entering feature construction.

Each connectivity matrix is vectorised into 19,900 pairwise features and standardised using training-partition statistics. Separate ℓ_2-regularised logistic-regression models are then fitted to the three representations, with the inverse regularisation parameter C selected by inner three-fold cross-validation. The resulting

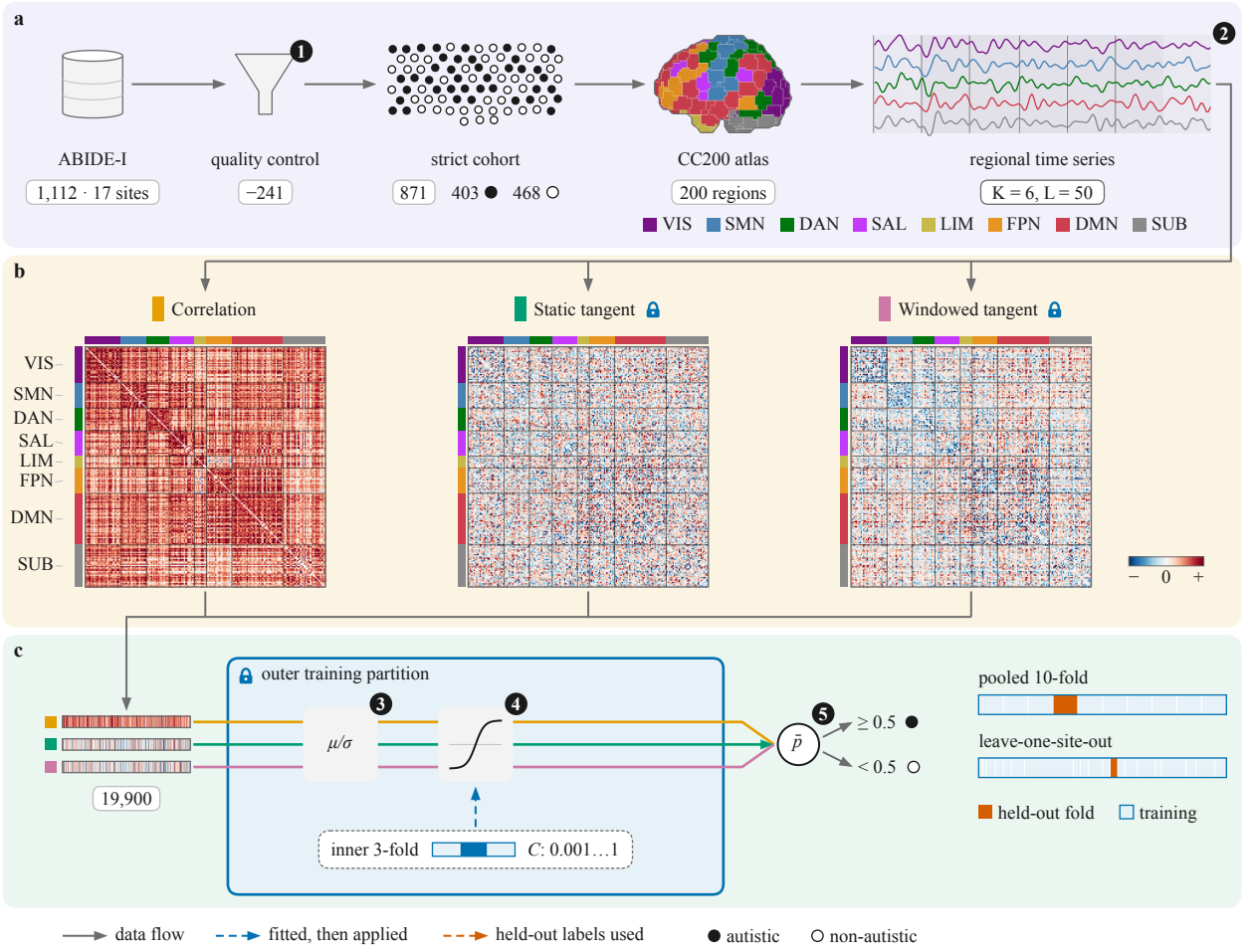


Figure 1: Overview of the primary analysis pipeline. (a) ABIDE-I participants undergo image-quality control to define the primary cohort, followed by extraction of CC200 regional time series from the preprocessed fMRI data. (b) Three functional-connectivity representations are derived for each participant: shrinkage-based correlation, static tangent-space connectivity, and windowed tangent-space connectivity. Tangent-space reference matrices are estimated from the corresponding training partition only. (c) Each representation is vectorised and standardised using training-derived statistics before fitting a separate ℓ_2-regularised logistic-regression model. The regularisation parameter is selected by inner three-fold cross-validation, and the three model probabilities are averaged to obtain the final ensemble prediction. Generalisation is evaluated using pooled ten-fold and leave-one-site-out cross-validation. Circled numbers indicate the principal stages of the pipeline.

probabilities are averaged with equal weights to produce the final ensemble prediction. Generalisation is evaluated using both pooled ten-fold cross-validation and leave-one-site-out validation across the 20 site labels. Throughout the primary pipeline, all data-dependent transformations and model parameters are estimated within the outer training partition before being applied to held-out participants, thereby maintaining separation between model fitting and evaluation.

The Methods are organised into nine subsections that follow the analysis pipeline. Section 3.1 describes the study design, data, cohorts, phenotypic variables, atlas, and scan-quality characteristics. Sections 3.2–3.5 then define the connectivity representations, prediction model, validation strategy, and probability-based analyses. The planned comparisons and exploratory group-contrast analyses are presented in Sections 3.6 and 3.7, followed by the statistical analysis in Section 3.8. Reproducibility and the scope of the resulting estimates are addressed in Section 3.9.

3.1. Study design and cohort

This subsection defines the study question, the data, and the participant cohorts on which the analyses are based. These definitions are given first, because all representations, models, and comparisons are based on these data and cohorts. The study design is described in Section 3.1.1, and the data source and preprocessing in Section 3.1.2. The primary cohort is then defined in Section 3.1.3, and the wider quality-control cohort in Section 3.1.3. Next, the phenotypic variables and the handling of missing values are specified in Section 3.1.4. The atlas and the signal-quality procedure are described in Section 3.1.5. Finally, the treatment of differences in scan length is described in Section 3.1.6.

3.1.1. Study design

Clinical diagnosis of autism spectrum disorder (ASD) combines direct observation, developmental history, and professional judgement. In this study, ASD status is defined by the diagnostic labels provided in the Autism Brain Imaging Data Exchange I (ABIDE-I), while resting-state functional magnetic resonance imaging (rs-fMRI) is used to characterise large-scale functional connectivity in autistic and non-autistic participants.

The ASD classification target is derived directly from the ABIDE-I diagnostic labels. Participants with `DX_GROUP = 1` are coded as autistic, whereas those with `DX_GROUP = 2` are coded as non-autistic. The primary analysis therefore evaluates how well the imaging-based models distinguish autistic from non-autistic participants within the ABIDE-I research cohort.

Discrimination is evaluated primarily using participant-pooled out-of-fold area under the receiver operating characteristic curve (AUROC) under pooled ten-fold and leave-one-site-out validation. For each participant, the prediction used for evaluation is generated by a model fitted without that participant's held-out data, and predictions from all held-out folds are subsequently combined to calculate the overall AUROC. AUROC is used as the primary discrimination measure because it assesses the ranking of autistic and non-autistic participants across prediction scores without requiring a fixed classification threshold.

In pooled ten-fold validation, participants from all sites are distributed across ten folds, with each fold held out once, providing an estimate of generalisation within the combined ABIDE-I multisite cohort. Leave-one-site-out validation instead holds out all participants associated with one site label at a time, providing a more stringent assessment of generalisation to an ABIDE-I site that is absent from model training.

The primary prediction model is an equal-weight ensemble of three connectivity representations. Beyond this model, six comparative analyses address the main contributions of the study. Four of these examine how discrimination varies with cognitive ability, whether functional-connectivity differences persist after matching for head motion, how feature-selection scope and granularity influence predictive performance, and how different classifier families perform under a common representation and validation procedure. The remaining two concern the symptom measures and the windowed description. For the symptom measures, associations with classifier scores are examined both across the full cohort and within autistic participants. For the windowed description, the whole-scan description is given the same covariance shrinkage, so that the two are compared under matched regularisation. Figure 3b–e summarises the designs of the first four comparative analyses, including the corresponding matching, feature-selection, subgroup, and classifier-comparison procedures.

3.1.2. Data source and preprocessing

Data are obtained from the Autism Brain Imaging Data Exchange I (ABIDE-I), a multi-site research dataset that combines resting-state functional MRI (rs-fMRI), anatomical MRI, and phenotypic information from 1,112 participants across 17 acquisition sites [19]. The phenotypic records provide diagnostic status together with demographic and cognitive measures, including age, sex, full-scale IQ, verbal IQ, performance IQ, handedness, and acquisition site. These variables are retained to characterise the study population and to support the subgroup and covariate analyses described later. After the quality-control procedure described in Section 3.1.3, the primary cohort contains 871 participants, including 403 autistic and 468 non-autistic participants. The cohort includes 727 male and 144 female participants and spans 6.5–58 years of age, with a mean age of 16.9 years (SD 7.6). Mean full-scale IQ is 105.8 in the autistic group and 111.1 in the non-autistic group. Because ABIDE-I combines data acquired independently at multiple centres, heterogeneity remains across sites in scanner hardware, acquisition protocols, scan duration, participant composition, and image

quality [1]. Image quality is therefore considered explicitly during cohort selection using the anatomical and functional quality ratings provided with the release. Preprocessed imaging outputs are obtained from the Preprocessed Connectomes Project [12], which provides ABIDE-I data processed through standardised workflows and thereby supports consistent preprocessing across participants from different acquisition sites. The use of these publicly released derivatives also improves reproducibility by fixing the imaging inputs and preprocessing choices. The present study uses outputs generated with the Configurable Pipeline for the Analysis of Connectomes (C-PAC), specifically the `filt_noglobal` processing strategy.

Within the selected C-PAC workflow, temporal and spatial sources of unwanted variation are addressed before functional connectivity is estimated. Slice-acquisition timing differences are corrected, successive functional volumes are realigned to reduce motion-related displacement, non-brain tissue is removed, signal intensities are normalised, and the functional images are aligned with the corresponding anatomical images. Additional variation associated with head motion and non-neural signals is reduced through nuisance regression using a 24-parameter motion model, five CompCor components derived from white-matter and cerebrospinal-fluid signals, and linear and quadratic temporal trends. This processing reduces contributions from residual head motion, physiological noise, and slow signal drift before connectivity is calculated. The resulting signals are then band-pass filtered between 0.01 and 0.10 Hz to retain the low-frequency fluctuations used for resting-state functional-connectivity analysis, while global signal regression is not applied under the selected `filt_noglobal` strategy [12]. Following preprocessing, the functional data are reduced from voxel-level images to atlas-defined regional time series. For the primary analysis, the CC200 atlas divides the brain into 200 functionally defined regions, and the signal within each region is summarised across the scan to produce one time series per region and participant. These 200 regional time series form the common imaging input from which the correlation, static tangent-space, and windowed tangent-space connectivity representations are constructed. Although the common C-PAC workflow reduces variation caused by inconsistent preprocessing procedures, heterogeneity arising from acquisition conditions and participant composition remains part of the ABIDE-I multisite setting. These remaining sources of variation are addressed through quality-controlled cohort selection, site-aware validation, nuisance adjustment, and the sensitivity analyses described in the following sections.

3.1.3. Primary and wider quality-control cohorts

The analysis cohorts are defined using the anatomical and functional image-quality ratings provided with ABIDE-I. These ratings identify whether the available scans are suitable for analysis and are accompanied by annotations for issues such as excessive head motion, unsuccessful skull stripping, incomplete or distorted anatomical coverage, preprocessing failure, image noise, missing imaging modalities, and regional signal dropout. For the primary cohort, a participant is retained only when the required assessment from all three raters is satisfied: rater 1 must mark the scan as acceptable; rater 2 must rate both the anatomical and functional images as acceptable or borderline; and rater 3 must rate both images as acceptable. Applying this criterion excludes 241 of the 1,112 ABIDE-I records. Because the quality annotations can overlap, an excluded scan may be associated with more than one recorded issue. The resulting primary cohort contains 871 participants, including 403 autistic and 468 non-autistic participants. It comprises 727 male and 144 female participants, with ages ranging from 6.5 to 58 years (mean 16.9 years; median 14.8 years). Cohort membership is fixed before the primary analyses, and participant identifiers are cross-checked against the phenotypic records to ensure that the same participants are used throughout the prediction pipeline. The cohort-selection procedure and subsequent time-series extraction are summarised in Figure 1a.

Participant characteristics by diagnostic group are reported in Table 1. Welch's t-test is used to compare age and the three intelligence quotient measures—full-scale, verbal, and performance IQ—between the autistic and non-autistic groups, allowing for unequal group variances. Mean framewise displacement (FD), defined in Section 3.1.4, is examined using both Welch's t-test and the Mann–Whitney U test. The former compares group means, whereas the latter provides a rank-based comparison that is less sensitive to the right-skewed FD distribution. Scan length is compared using the Mann–Whitney U test, and sex distributions are compared using the χ^2 test with Yates' continuity correction. Because cognitive ability and head motion may influence subsequent results, their effects are examined separately in the cognitive-subgroup analysis of Section 3.6.3 and the motion-matched group-contrast analysis of Section 3.7.3.

Table 1
Participant characteristics in the primary cohort (871 participants; 20 site labels from 17 acquisition sites). Values are mean (SD) unless stated; the number of participants with an available value is given in brackets where it is below the group total.

Characteristic	Autistic (n = 403)	Non-autistic (n = 468)	p
Age, years	17.07 (7.96)	16.84 (7.24)	0.66
Female, n (%)	54 (13.4)	90 (19.2)	0.027
Full-scale IQ	105.79 (17.06) [376]	111.09 (12.08) [435]	6.2×10^{-7}
Verbal IQ	104.89 (18.10) [342]	111.57 (13.07) [379]	2.8×10^{-8}
Performance IQ	105.69 (17.26) [344]	107.91 (13.05) [391]	0.052
Mean framewise displacement (Section 3.1.4), mm	0.126 (0.143)	0.094 (0.082)	8.2×10^{-5}
Mean framewise displacement (Section 3.1.4), mm, median [IQR]	0.088 [0.057, 0.144]	0.068 [0.045, 0.109]	8.9×10^{-7}
Volumes per scan, median [range]	176 [78, 316]	176 [78, 316]	0.77

p-values are from Welch's t-test for rows summarised by mean (SD), the Mann–Whitney U test for rows summarised by median, and the χ^2 test with Yates' continuity correction for sex.

A wider quality-control cohort is additionally defined to test whether the findings depend on the more restrictive primary inclusion rule and to support comparison with previous ABIDE-I studies. For this cohort, eligibility requires an acceptable functional-image rating from rater 3, without applying the full three-rater criterion used for the primary cohort. This definition identifies 1,035 eligible participants in the phenotypic records, of whom 1,032 have regional time series available for analysis. The 1,035-participant eligibility count matches the cohort definition reported by Heinsfeld et al. [24], Eslami et al. [20], and Almuqhim and Saeed [3]. The wider cohort is therefore used as a cohort-definition sensitivity analysis, allowing the primary results to be assessed under a less restrictive quality-control rule while also providing a direct basis for comparison with these previous reports.

3.1.4. Phenotypic variables

Phenotypic information is linked to each participant through the ABIDE-I identifiers and includes age, sex, full-scale IQ, verbal IQ, performance IQ, handedness, and acquisition site. These variables are used to characterise the cohort and to support the secondary, subgroup, and nuisance-adjustment analyses described later. The primary ensemble itself is restricted to imaging-derived connectivity features, so demographic and cognitive variables do not contribute directly to its predictions. A separate phenotype model, described in Section 3.6.2, instead uses age, sex, the three IQ measures, and handedness to evaluate discrimination from non-imaging information. Acquisition site is retained for site-based validation, nuisance adjustment, and sensitivity analyses. Head motion is represented separately by mean framewise displacement (FD), obtained from the `func_mean_fd` quality metric provided with the preprocessed release. This measure follows Jenkinson's formulation and is calculated from the rigid-body realignment parameters between successive functional volumes, with rotational components converted to millimetres using an 80-mm head radius [30]. The resulting mean FD provides a single scan-level summary of head motion for each participant and is used in the participant-characteristic comparisons and the subsequent motion-adjusted and motion-matched analyses.

Missing phenotypic entries are identified before analysis, including values recorded as `-9999` or left blank. When phenotypic variables are used in predictive or harmonisation models, missing continuous values are replaced by the median calculated from the corresponding outer training partition. This prevents information from held-out participants from entering preprocessing. The same rule is used for the secondary phenotype model in Section 3.6.2 and for age in the ComBat analysis of Section 3.6.2. In the ComBat model, sex is coded as 0 for male and 1 for female. No age or sex values are missing in the primary cohort. For analyses fitted directly to a complete analysis cohort, required missing covariates are median-imputed from that cohort. This applies to the nuisance-adjustment procedures in Sections 3.6.1 and 3.7.1, as well as to the preparation for motion matching in Section 3.7.3. In contrast, no imputation is applied in the symptom-association,

cognitive-subgroup, instrument-concordance, component–symptom, or participant-characteristic analyses. Participants with missing values are excluded only from the analysis that requires the corresponding variable.

3.1.5. Atlas and signal quality

The primary analyses use the Craddock 200-region functional parcellation (CC200) [13]. All 200 regions are retained for prediction, yielding 19,900 unique pairwise connectivity features. A separate signal-quality check is applied before the group-difference analyses because Pearson correlation cannot be computed for a region with no temporal variation. First, any region with a temporal standard deviation of zero in more than 1% of the 1,032 available CC200 time-series files is removed; four regions meet this criterion. Participants in the primary cohort are then excluded from the group-contrast analysis if any of the remaining regions contains a constant signal. This produces a group-contrast cohort of 863 participants and 196 regions, corresponding to 19,110 unique connections. To support system-level interpretation, each CC200 region is also assigned an anatomical label and a functional-system label. Yeo's seven cortical systems are mapped onto the CC200 space using nearest-neighbour interpolation [57], preserving a single system label for each voxel. Each region is assigned to the system with the greatest spatial overlap when that system covers more than 20% of the region; regions that do not meet this threshold are assigned to a subcortical/cerebellar category. This procedure yields eight functional-system classes and 36 unique within- and between-system combinations, which are used to summarise connection-level group effects at the system level.

3.1.6. Scan length

Scan length is considered explicitly in the dynamic-connectivity analyses because the number of available volumes differs substantially across participants and sites. In the primary cohort, scans contain between 78 and 316 volumes, with scan duration varying systematically by acquisition site. To keep the dynamic representations comparable, each participant contributes the same number of temporal windows regardless of total scan length. This prevents participants with longer scans from contributing more window-level estimates than those with shorter scans. Window length and temporal change are expressed in volumes rather than seconds because a harmonised repetition-time field is not available across the analysis files.

3.2. Connectivity representations

This subsection describes how each participant's regional rs-fMRI time series are transformed into the three connectivity representations used by the primary ensemble: shrinkage correlation, static tangent-space connectivity, and windowed tangent-space connectivity. The first two describe connectivity over the full scan, whereas the third summarises connectivity estimated from shorter temporal windows. Each representation is converted into a 19,900-dimensional feature vector for the 200-region CC200 atlas and is processed independently before classification. The corresponding feature construction, covariance estimation, tangent-space projection, additional shrinkage, and windowing procedures are detailed in Sections 3.2.1–3.2.6.

Figure 2 summarises this transformation. Whole-scan and window-level covariance matrices are first estimated using Ledoit–Wolf shrinkage. Static tangent connectivity is obtained by projecting the whole-scan covariance relative to a Riemannian reference estimated from the corresponding training partition, while shrinkage correlation is derived directly from the covariance estimate. For the windowed representation, each scan is divided into six windows of 50 volumes, and the window covariances receive an additional isotropic shrinkage of $r = 0.15$ before projection to a training-derived dynamic reference. The resulting window tangent matrices are averaged to form the mean windowed tangent representation. All training-dependent reference matrices are re-estimated within each outer fold and then applied unchanged to held-out participants.

3.2.1. Functional-connectivity features

Functional-connectivity features are derived from the regional rs-fMRI time series of each participant. Each CC200 region is represented by a blood-oxygen-level-dependent signal recorded across the scan, and connectivity is estimated from the statistical relationship between pairs of these regional signals. With 200 regions, this produces 19,900 unique pairwise connectivity features for each participant. Three representations of these connections are used in the primary ensemble: shrinkage correlation, static tangent-space connectivity, and mean windowed tangent-space connectivity. The static tangent representation summarises connectivity across the full scan, whereas the windowed representation captures the average

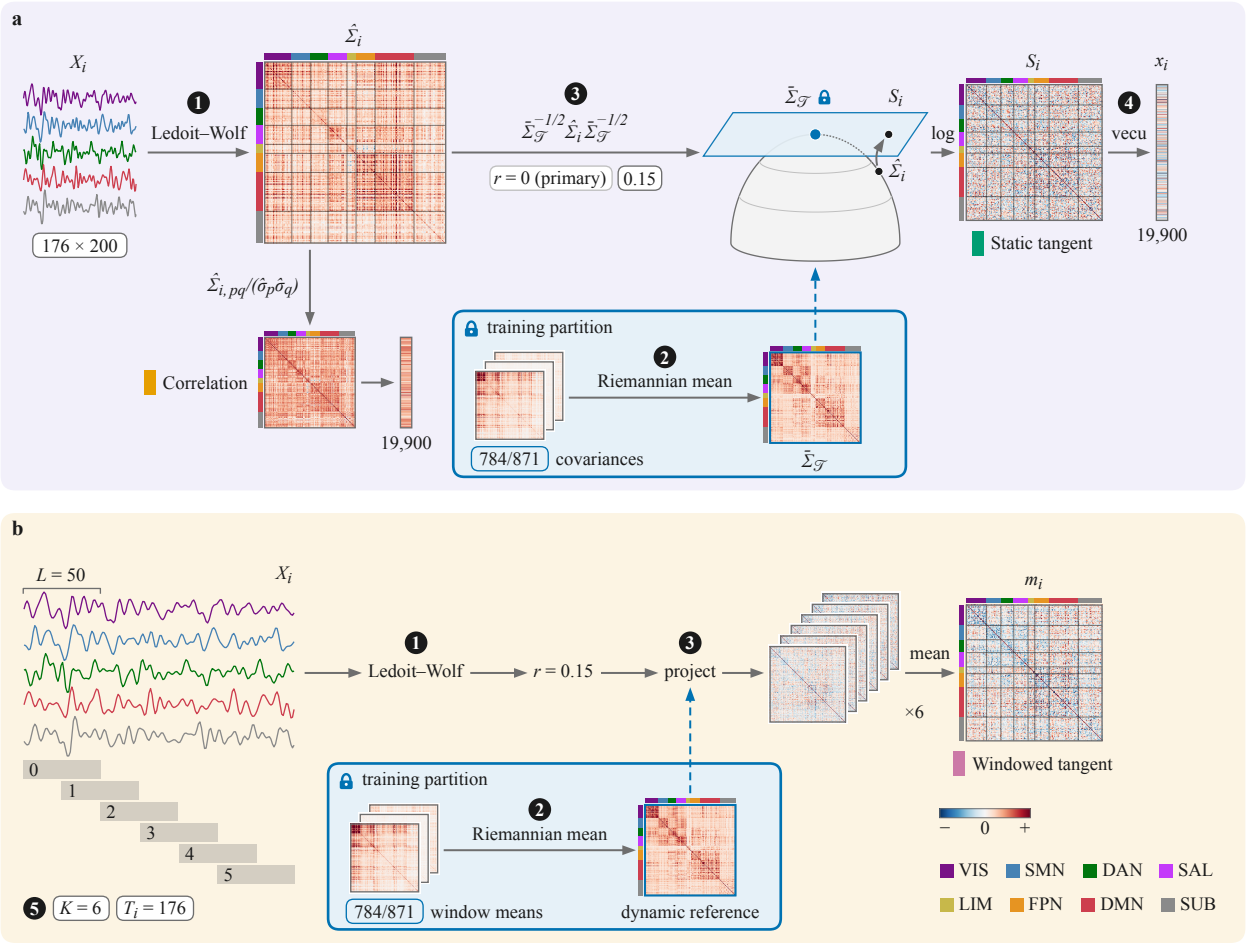


Figure 2: Construction of the connectivity representations for one example participant. (a) 1 Ledoit–Wolf covariance, 2 training-only Riemannian reference, 3 whitening and matrix logarithm, and 4 vectorisation; a shrinkage-correlation branch is also shown. (b) 5 Six windows of 50 volumes, each processed by steps 1–3 at a dynamic reference and then averaged. Blue lanes mark training-only quantities.

connectivity pattern across shorter temporal segments. These three representations are illustrated for one participant in Figure 1b. Each representation is constructed and standardised separately so that its fitted reference and scaling parameters remain specific to that representation.

3.2.2. Shrinkage-correlation features

Shrinkage-correlation features are derived from each participant's Ledoit–Wolf covariance estimate, defined in Section 3.2.3. This estimate provides a regularised description of how the regional signals vary together and is converted to a correlation matrix by normalising each covariance by the standard deviations of the corresponding region pair. The resulting coefficients therefore describe pairwise coupling on a common scale, independent of regional signal variance. The strict upper triangle of the correlation matrix is then vectorised into a single feature vector, while the diagonal is omitted because all self-correlations equal one. This produces 19,900 correlation features for the CC200 atlas. The same outer-fold partitions are used for this representation and the two tangent-space representations, ensuring that all three ensemble members are trained and evaluated on identical participant splits.

3.2.3. Whole-scan covariance estimation

Whole-scan covariance is estimated for each participant from all volumes of the regional time series. For this purpose, $X_i \in \mathbb{R}^{T_i \times P}$ denotes the regional time series for participant i, where T_i is the number of volumes and P is the number of regions. The covariance is then estimated with Ledoit–Wolf shrinkage [33], as given in Equation (1). In this estimator, the covariance is a weighted average of the sample covariance and a scaled identity matrix:

$$\widehat{\Sigma}_i = (1-\rho_i)\widehat{S}_i + \rho_i \widehat{\mu}_i I_P, \qquad \widehat{\mu}_i = \frac{\operatorname{tr}(\widehat{S}_i)}{P}. \tag{1}$$

In Equation (1), $\widehat{S}_i$ is the sample covariance, I_P is the identity matrix, and $\widehat{\mu}_i$ is the average of the regional variances. The shrinkage weight ρ_i is, in turn, selected by the Ledoit–Wolf estimator. Shrinkage is used because the number of regions is large relative to the number of volumes, which makes the sample covariance alone an unstable estimate.

3.2.4. Tangent-space projection

Tangent-space projection converts each covariance matrix into a representation that can be used directly by standard linear classifiers. Because covariance matrices have a non-Euclidean structure, they are projected relative to a reference matrix estimated from the training data. This produces a locally linear feature space while preserving the geometry of the original covariance matrices. The projection process is illustrated in Figure 2a.

The reference matrix is estimated separately for each outer fold as the affine-invariant Riemannian mean of the training covariances [14, 54]. As given in Equation (2), this mean is the positive-definite matrix R with the smallest summed squared distance to the training covariances:

$$\overline{\Sigma}_{\mathcal{T}} = \arg\min_{R \succ 0} \sum_{i \in \mathcal{T}} \left\| \log\left(R^{-1/2} \widehat{\Sigma}_i R^{-1/2} \right) \right\|_F^2. \tag{2}$$

In Equation (2), $\mathcal{T}$ denotes the training set, and the Frobenius norm is used to quantify the affine-invariant distance between the reference matrix R and each training covariance. The Riemannian mean is therefore adopted as the reference because it respects the geometry of the covariance space. It is estimated by fixed-point iteration with at most ten updates. Matrix logarithms and inverse square roots are computed by symmetric eigendecomposition, with eigenvalues clamped at 10^{-6} to maintain numerical stability.

Training and held-out covariances are then projected into the tangent space with the training reference. As shown in Equation (3), each covariance is first multiplied on both sides by the inverse square root of the reference. The matrix logarithm of this product then gives the projected matrix S_i:

$$S_i = \log\left(\overline{\Sigma}_{\mathcal{T}}^{-1/2} \widehat{\Sigma}_i \overline{\Sigma}_{\mathcal{T}}^{-1/2} \right). \tag{3}$$

The projected matrix in Equation (3) is then vectorised by retaining only its off-diagonal upper triangle, as given in Equation (4). Only the upper triangle is retained because the matrix is symmetric, so the entries below the diagonal repeat those above it. Here, vecu stacks the entries above the diagonal into a vector:

$$x_i = \operatorname{vecu}(S_i) \in \mathbb{R}^{P(P-1)/2}. \tag{4}$$

For CC200, the vector in Equation (4) therefore contains 19,900 features. In addition, the reference matrix is estimated from the training partition and applied unchanged to held-out participants. As a result, no information from held-out participants enters the tangent projection.

3.2.5. Additional isotropic shrinkage

Additional isotropic shrinkage is used in the planned comparisons to further regularise the Ledoit–Wolf covariance estimates. The primary static tangent representation uses the Ledoit–Wolf estimate directly, corresponding to $r = 0$. For comparison with the windowed representation, an additional shrinkage level of $r = 0.15$ is applied using Equation (5):

$$\widehat{\Sigma}_i(r) = (1-r)\widehat{\Sigma}_i + r\frac{\operatorname{tr}(\widehat{\Sigma}_i)}{P} I_P, \qquad 0 \le r < 1. \tag{5}$$

A separate nested sensitivity analysis selects r from $\{0, 0.10, 0.20, 0.30, 0.40, 0.50\}$ within each outer training partition. Classifier regularisation is reselected for every candidate value of r, so each shrinkage setting is evaluated with its corresponding tuned classifier while the held-out data remain excluded from parameter selection.

3.2.6. Windowed tangent-space representation

The windowed tangent-space representation is built from covariance matrices estimated within the windows of each scan. In this approach, window-level covariance matrices are used to represent dynamic connectivity [2, 28, 34, 47]. From these matrices, the mean windowed tangent representation, hereafter termed the dynamic mean, forms the third member of the primary ensemble. In contrast, dispersion, velocity, and alternative block combinations, that is, combinations of the mean, dispersion, and velocity blocks, are secondary analyses. Dispersion and velocity are, in turn, defined below. The placement of the six windows and the formation of the dynamic mean from the window tangent matrices are illustrated in Figure 2b.

For every participant, $K = 6$ windows of $L = 50$ volumes are extracted. The values $K = 6$ and $L = 50$ are used because every scan contains at least 78 volumes, so that each window spans 50 volumes. The window positions are given in Equation (6), which spaces the windows evenly from the start to the end of each scan. Specifically, for a scan with T_i volumes, Δ_i is the nominal stride between consecutive window starts, and s_{ik} is the start of window k:

$$\Delta_i = \frac{T_i - L}{K - 1}, \qquad s_{ik} = \mathrm{round}(k\Delta_i), \qquad k = 0, \ldots, K - 1. \tag{6}$$

The number of windows is fixed across participants to keep the dynamic summaries comparable despite differences in scan length across sites. Shorter scans therefore contain more overlap between windows, while longer scans may contain small gaps, and each window start is rounded to the nearest acquired volume. For every window, covariance is estimated using Ledoit–Wolf shrinkage with an additional isotropic shrinkage of $r = 0.15$ to improve numerical stability. Adaptive diagonal jitter is applied when required during eigendecomposition. The dynamic training reference is then constructed by first averaging the six window covariances for each training participant and subsequently taking the Riemannian mean across participants. This procedure gives each training participant equal weight in the reference.

Three summaries are then computed from the projected windows of each participant. After tangent projection, participant i is represented by the window vectors $\xi_i(1), \ldots, \xi_i(K)$, which form a trajectory across the scan. Each summary is computed elementwise, that is, separately for every connection. Specifically, the mean m_i, the dispersion d_i, and the velocity v_i are given in Equations (7), (8), and (9), respectively:

$$m_i = \frac{1}{K} \sum_{k=1}^{K} \xi_i(k), \tag{7}$$

$$d_i = \left[\frac{1}{K} \sum_{k=1}^{K} \left(\xi_i(k) - m_i \right)^2 \right]^{1/2}, \tag{8}$$

$$v_i = \frac{1}{(K-1)\Delta_i} \sum_{k=1}^{K-1} \left| \xi_i(k+1) - \xi_i(k) \right|. \tag{9}$$

The three summaries capture complementary properties of the window trajectory. The mean in Equation (7) represents the average connectivity pattern across windows, while the dispersion in Equation (8) quantifies within-scan variability around this mean using the denominator K. The velocity in Equation (9) measures the average absolute change between consecutive windows. It is normalised by the stride from Equation (6), allowing temporal change to be compared across scans with different window spacing.

3.3. Prediction model

This subsection describes how the three connectivity representations are converted into participant-level predictions. Each representation is standardised and classified separately using regularised logistic regression, after which the three probabilities are combined to form the primary ensemble prediction. Sections 3.3.1–3.3.3 describe scaling, classification, and probability averaging, while Section 3.3.4 presents the secondary rank-fusion analysis.

3.3.1. Feature scaling

Before classification, each feature block, that is, the features of one representation, is standardised. To this end, every feature is centred on its mean and divided by its standard deviation. Both statistics are estimated from the outer training partition, as shown in Equation (10):

$$z_{ij} = \frac{x_{ij} - \mu_{j,\mathcal{T}}}{\sigma_{j,\mathcal{T}}}. \tag{10}$$

The per-representation scaling, the regularised logistic models, and the probability ensemble are summarised in Figure 1c.

In Equation (10), x_{ij} is feature j of participant i, and $\mu_{j,\mathcal{T}}$ and $\sigma_{j,\mathcal{T}}$ are the mean and standard deviation of that feature over the training indices $\mathcal{T}$. The transformation is fitted on the training partition and applied unchanged to the held-out partition. Because both statistics come from the training partition alone, no information from held-out participants enters the scaling. Standardisation, in turn, places all features on a common scale before the penalised classifier is fitted. In addition, separate scalers are fitted for the correlation, static tangent, and dynamic tangent members, so that each member is standardised with its own training statistics.

3.3.2. Regularised logistic regression

Logistic regression with an ℓ_2 penalty is used for each primary member. Logistic regression is a linear model that converts a weighted sum of the features into a probability. The ℓ_2 penalty, in turn, shrinks the weights towards zero. The penalty is used because the number of connectivity features exceeds the number of training participants. For participant i, the predicted probability is given by Equation (11):

$$p_i = \sigma(w^\top z_i + b), \qquad \sigma(a) = \frac{1}{1 + \exp(-a)}. \tag{11}$$

In Equation (11), w and b denote the model weights and intercept, z_i is the standardised feature vector, and σ is the logistic function. Logistic regression is fitted in scikit-learn [45] using the `liblinear` solver with a maximum of 2,000 iterations. The regularisation parameter C is selected separately for each representation from $\{0.001, 0.01, 0.1, 1.0\}$ using three-fold stratified cross-validation within the outer training partition, with AUROC used for selection. Class weights are kept uniform. For the primary ensemble, the tangent reference and feature scaler are fitted once on the complete outer training partition and reused during the inner search for C. The outer held-out partition remains excluded from both transformation fitting and model selection.

3.3.3. Probability ensemble

In the primary model, the probabilities from whole-scan correlation, whole-scan tangent connectivity, and mean windowed tangent connectivity are averaged. For member probabilities $p_i^{(1)}, p_i^{(2)}, p_i^{(3)}$, the ensemble probability is given by Equation (12):

$$p_i^{\text{ens}} = \frac{1}{3} \sum_{m=1}^{3} p_i^{(m)}. \tag{12}$$

The members in Equation (12) are given equal weight. Moreover, ensemble membership and weighting are fixed across outer folds. Equal weighting is used because it avoids estimation of additional fusion parameters from the training sample. Finally, the ensemble probability is converted into a class decision with a default class threshold of 0.5.

3.3.4. Secondary rank fusion

As a secondary analysis, the three model outputs are combined using their relative ranks rather than their predicted probabilities. For each held-out participant, a member score is located within the distribution of scores produced for that member's training participants. This converts the score to a value between 0 and 1 that reflects its relative position within the training distribution. When scores are tied, the held-out value is placed after the equal training scores. The three rank-based scores are then averaged to obtain the fused score. This approach places all ensemble members on a common scale while avoiding dependence on their original probability ranges. The fused value is therefore interpreted as an ordering score, not as a probability, and each held-out participant is ranked only against training scores.

3.4. Validation protocol

[THIS PART NEEDS REFERENCE CHECKING]

Two outer validation schemes are used to assess model performance: pooled ten-fold cross-validation and leave-one-site-out validation. Within each outer training partition, classifier regularisation is selected through inner cross-validation, resulting in a nested evaluation design. All training-dependent transformations and model parameters are estimated from the outer training data only, keeping the held-out partition separate from model fitting and model selection. Outer validation and fold-level fitting is discussed in Section 3.4.1. Performance measures and fold-level aggregation are presented in Section 3.4.2, while interval estimation is described in Section 3.4.3.

3.4.1. Outer validation and fold-level fitting

Model evaluation is performed using both pooled ten-fold cross-validation and leave-one-site-out validation. In pooled ten-fold validation, the primary cohort is divided into ten stratified folds with similar diagnostic proportions, using shuffling with random seed 42. Each fold is held out once, so every participant contributes one test prediction and performance reflects generalisation within the combined ABIDE-I site mixture. In leave-one-site-out validation, each of the 20 recorded site labels is held out in turn, while participants from the remaining sites are used for training. This provides a stricter assessment of transfer to an ABIDE-I site that is absent from model fitting. Within every outer training partition, stratified inner folds are used to select the regularisation parameter C. Model fitting then follows the same sequence for both validation schemes: training and held-out indices are first defined, the tangent reference is estimated from the training covariances, and both partitions are projected using this fixed reference. Feature scaling is subsequently fitted on the training features and applied unchanged to the held-out data before classification. This procedure keeps all fitted transformations and model-selection steps within the training partition. The two outer validation schemes are summarised in Figure 1c, while the nested fitting procedure is illustrated in Figure 3a.

Figure 3 summarises the main validation and comparative designs used in the study. Panel (a) shows the nested fitting procedure: Ledoit–Wolf covariances are computed before the outer split, while tangent references, scaling parameters, and classifier tuning are restricted to the outer training partition. The selected classifier is then refitted, and the three member probabilities are averaged to form the ensemble prediction. Panel (b) compares connection- and region-level feature selection, including both training-fold selection and deliberately global label-informed selection. Panel (c) presents the group-contrast analysis, where connectivity effects are estimated in the full cohort and in within-site samples matched on framewise displacement, followed by within-site permutation testing and BH-FDR correction. Panel (d) summarises the cognitive analyses, including comparisons across autistic IQ bands, matched comparisons with the non-autistic group, and within-group score–IQ associations. Panel (e) compares linear and tree-based classifiers under the same static tangent representation, scaling procedure, and outer folds, with only the ℓ_2-regularised logistic model receiving hyperparameter tuning. The arrows and colour conventions follow Figure 1, while the remaining markers and matrices are schematic.

The classifier is then tuned and refitted on the transformed training data. To this end, C is selected by inner cross-validation on the transformed training matrix. The selected classifier is then refitted to the complete outer training partition, so that the final model uses every training participant. Next, probabilities are predicted for the held-out partition. Finally, the three member probabilities are combined.

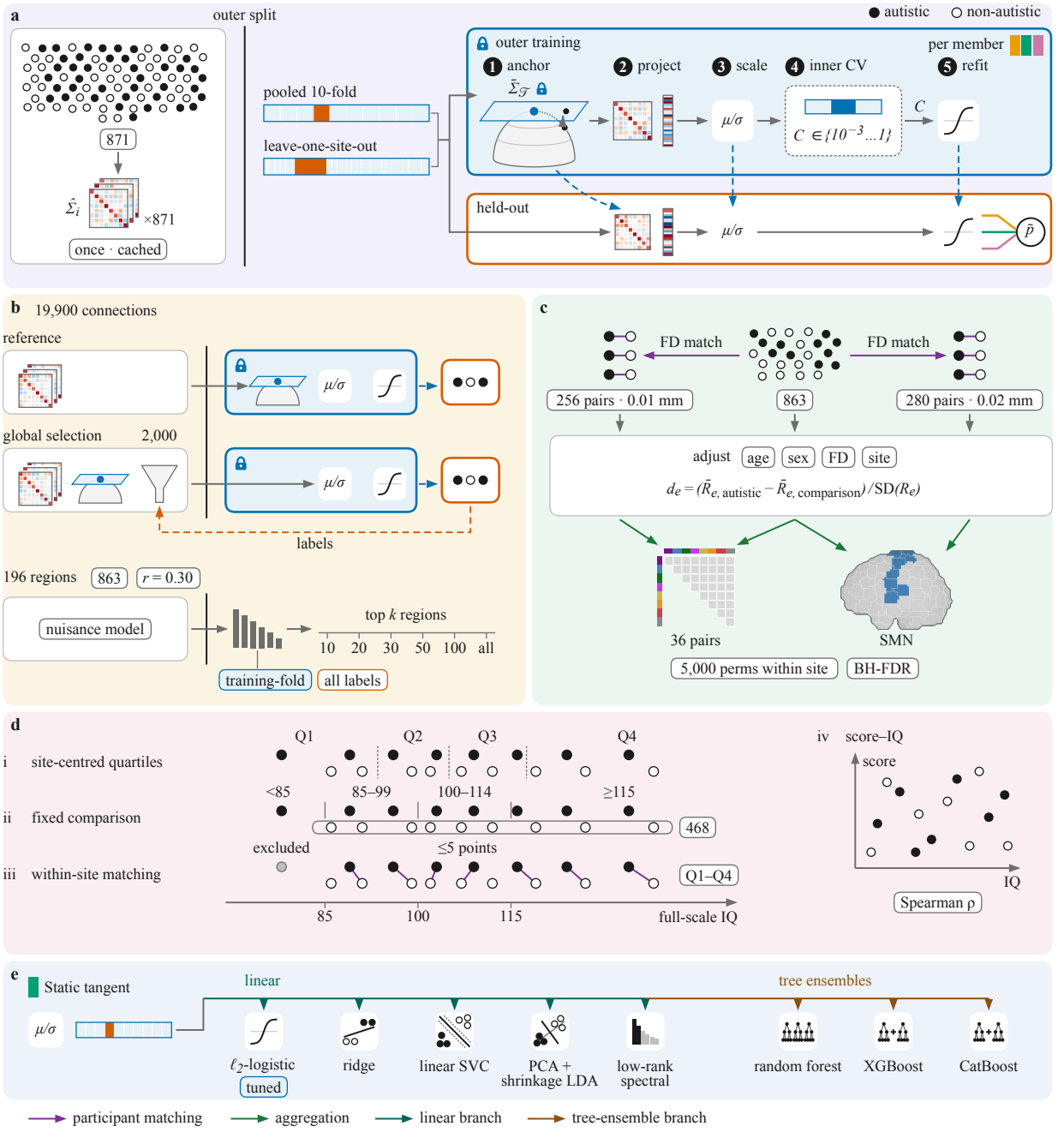


Figure 3: Evaluation and analysis designs. (a) Nested cross-validation in the outer training partition: ❶ reference, ❷ projection, ❸ scaling, ❹ inner-CV tuning, ❺ refitting. (b) Feature selection within training folds versus with all labels. (c) Motion-matched group contrast. (d) IQ designs: (i) quartiles, (ii) fixed comparison, (iii) matching, (iv) score–IQ correlation. (e) Classifier families.

Per-participant covariance estimation is handled separately from these fold-specific steps. Each covariance estimate depends only on that participant's time series. Therefore, it is cached before fold-specific fitting without using information from other participants. Shared quantities, in contrast, are estimated from several participants. For this reason, shared references, scalers, feature selectors, and classifier parameters are estimated at the training level specified for each analysis.

3.4.2. Performance measures and aggregation

Discrimination is evaluated primarily using AUROC and the area under the precision–recall curve, both calculated from continuous prediction scores. Threshold-based performance is also reported at 0.5 using accuracy, balanced accuracy, sensitivity, specificity, F1 score, and Matthews correlation coefficient, with autistic participants treated as the positive class. Balanced accuracy and Matthews correlation coefficient are included to provide measures that are less affected by unequal group sizes.

Performance is summarised in two ways across the outer folds. Participant-pooled metrics are calculated from all concatenated out-of-fold predictions, while unweighted fold means and standard deviations give equal importance to each fold. Both summaries are retained because site sizes vary substantially in leave-one-site-out validation. Planned comparison tables report fold-level AUROC and accuracy, whereas pooled out-of-fold predictions are used for calibration, paired prediction tests, and the main interval estimates.

3.4.3. Interval construction

Intervals are reported for the main performance measures. AUROC intervals are obtained from the DeLong variance, while Wilson intervals [56] are used for accuracy, sensitivity, and specificity. These intervals are conditional on the observed out-of-fold predictions and therefore do not include uncertainty from repeating the complete modelling procedure.

To summarise variation across acquisition sites, site-specific AUROCs from leave-one-site-out validation are combined using a DerSimonian–Laird random-effects model [17]. Only site labels containing both diagnostic groups are included, leaving J sites for analysis. Each site is weighted by $1/(V_j+\hat{\tau}^2)$, where V_j is its DeLong variance and $\hat{\tau}^2$ is the estimated between-site variance. Between-site heterogeneity is summarised using Cochran's Q and I^2, and the expected range of AUROC at a new site is estimated with the 95% prediction interval in Equation (13):

$$\hat{\mu} \pm t_{0.975,\,J-2}\sqrt{\hat{\tau}^2 + \mathrm{SE}(\hat{\mu})^2}. \tag{13}$$

Here, $\hat{\mu}$ is the random-effects mean AUROC and $\mathrm{SE}(\hat{\mu})$ is its standard error. By incorporating both uncertainty in the pooled mean and estimated between-site variation, the interval reflects the range of performance expected at an additional acquisition site.

3.5. Probability assessment and selective prediction

This subsection examines the reliability of the ensemble probabilities through calibration, risk–coverage analysis, and confidence-based selective prediction with finite-sample error control.

3.5.1. Secondary probability assessment

Beyond discrimination, the ensemble probabilities are evaluated for calibration and later used in selective prediction. The calibration analysis examines the native probabilities from Equation (12) without any additional recalibration. Their reliability is assessed by comparing predicted probabilities with the observed proportion of autistic participants. The probabilities are divided into ten equal-width bins, and the mean prediction in each bin is compared with the corresponding observed outcome frequency. These bin-level differences are then combined into the expected calibration error (ECE), weighted by the number of participants in each bin [23, 42], as defined in Equation (14):

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|\mathcal{B}_b|}{n} \left| \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} p_i^{\mathrm{ens}} - \frac{1}{|\mathcal{B}_b|} \sum_{i \in \mathcal{B}_b} y_i \right|. \tag{14}$$

In Equation (14), $\mathcal{B}_b$ denotes the set of participants in bin b, n denotes the total number of participants, and $y_i = 1$ denotes an autistic participant. In this sum, empty bins contribute no term. Because ECE is computed from the probability of the autistic class, it assesses positive-class probability calibration. In addition, the same bin-level quantities are displayed in reliability diagrams under both validation protocols. In such a diagram, the mean predicted probability of each bin is plotted against the observed proportion of autistic participants in that bin. These diagrams are included because they show in which probability range any disagreement occurs, whereas a single ECE value does not.

3.5.2. Exploratory risk–coverage analysis

In the exploratory risk–coverage analysis, the change in error rate is examined as predictions are retained only for the most confident participants. In this analysis, coverage is the fraction of participants whose predictions are retained. The analysis is performed because it shows whether higher model confidence corresponds to fewer incorrect decisions. To this end, prediction confidence is defined in Equation (15) as the larger of the ensemble probability and its complement:

$$c_i = \max\left(p_i^{\text{ens}}, 1 - p_i^{\text{ens}}\right). \tag{15}$$

Participants are then ordered by the confidence in Equation (15). Next, the number of selected participants, accuracy, AUROC, and selective risk are recorded at coverage levels from 10% to 100%. Here, selective risk is the fraction of incorrect decisions among the selected participants [22]. Specifically, at each requested coverage, the highest-confidence observations in the complete out-of-fold prediction set are retained. However, AUROC is undefined when a retained subset contains one diagnostic class, because AUROC requires participants from both classes.

3.5.3. Exploratory finite-sample threshold analysis

In the exploratory threshold analysis, a confidence threshold θ is selected from labelled calibration data. In this context, calibration data are participants with known diagnostic labels who are used to choose the threshold. For this selection, the candidate thresholds are the distinct observed confidence values, and a participant is selected when $c_i \geq \theta$. For each candidate θ, $k(\theta)$ denotes the number of selected participants, and $e(\theta)$ denotes the number of errors among them. The error rate of the selected participants is then bounded with the one-sided Clopper–Pearson upper bound [10], as given in Equation (16):

$$U(\theta; \delta) = \text{BetaInv}\left(1 - \delta; e(\theta) + 1, k(\theta) - e(\theta)\right). \tag{16}$$

In Equation (16), BetaInv is the beta-distribution quantile function and $1 - \delta$ sets the confidence level. The upper confidence bound is used instead of the observed error rate so that threshold selection accounts for sampling uncertainty in the calibration data. Candidate thresholds are evaluated in a fixed order, from the most restrictive to the most permissive. The search stops when the upper bound first exceeds the target risk α, and the last passing threshold is retained. Candidates selecting fewer than ten participants are skipped [4, 5]. The fixed ordering allows each candidate to be tested at level δ without an additional correction across thresholds.

Two boundary cases are handled explicitly. If the first eligible candidate already exceeds the target risk, the threshold is set to 1.01, which gives zero coverage because confidence scores cannot exceed one. If every selected case is an error, the upper bound is set to one because the beta quantile in Equation (16) is otherwise undefined. For the ensemble analysis, $\alpha = 0.25$, $\delta = 0.10$, and a minimum selected sample of ten are used, corresponding to a 25% target error with a 90% upper confidence bound. Two thresholding strategies are evaluated. Marginal thresholding uses one threshold for the full calibration sample, whereas site-conditional thresholding estimates a separate threshold for each site. When a site's calibration subset contains fewer than ten participants, the marginal threshold is used instead.

Threshold performance is assessed over 200 random, unstratified half-splits. In each repetition, one half of the data is used to select the threshold and the other half is used to measure realised coverage and selective risk. This keeps threshold evaluation separate from threshold selection. Repetitions with undefined selective risk are excluded from the summary. The reported results include the mean, median, and 90th-percentile risk, mean coverage, and the proportion of valid repetitions that satisfy the target risk. Repeated splitting is used to capture variation across different calibration samples rather than relying on a single partition.

3.6. Planned comparisons

This subsection presents the planned primary, secondary, and exploratory comparisons built on the same prediction and validation framework. These analyses examine how performance and interpretation vary with connectivity representation, classifier choice, atlas and feature-selection strategy, phenotypic and site effects, participant subgroups, data availability, and model interpretability.

3.6.1. Primary descriptive comparison

Static and windowed connectivity. The planned comparisons use the three-member prediction model under both pooled ten-fold and leave-one-site-out validation, with each analysis classified by its status in Section 3.8.1. The first primary descriptive comparison evaluates static tangent connectivity against the dynamic mean while controlling the level of additional shrinkage. Because the windowed representation uses $r = 0.15$, static tangent connectivity is evaluated at both $r = 0$ and $r = 0.15$, and the dynamic mean is compared directly with the $r = 0.15$ static representation. Other representation-specific steps, including window sampling, covariance aggregation, and reference construction, remain unchanged. Static tangent performance is also examined across the shrinkage grid $0, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40, 0.50, 0.65, 0.80$. The sweep is first evaluated up to 0.30 and is later extended to 0.80 to examine stronger shrinkage, without changing the fixed $r = 0.15$ comparison. In a separate nested analysis, the shrinkage level is selected within each outer training partition. The fixed and data-selected analyses are therefore reported separately, since one uses a predefined shrinkage level while the other estimates it from the training data.

Feature-selection scope. This primary descriptive comparison examines how model performance changes when fitted steps are allowed to access data beyond the outer training fold. Seven data-access conditions are evaluated using the 19,900 static tangent features from the primary cohort, while the same participants and outer folds are retained throughout. The reference condition fits the tangent reference, scaler, and classifier within each outer training fold. The global-reference condition estimates the tangent reference from all participants but retains fold-specific scaling, while the global-scaling condition fits both the reference and scaler globally. In the global-selection condition, the tangent reference is fitted globally, scaling remains fold-specific, and the 2,000 strongest connections are selected with an F test using all diagnostic labels. These features represent approximately 10% of all connections. A held-out-tuning condition selects C using the held-out fold, whereas a site-feature condition adds one-hot site indicators to the otherwise fold-specific pipeline. Finally, the combined condition applies global tangent projection, global scaling, global label-informed feature selection, held-out C selection, and site indicators together. Here, "global" denotes any step fitted using all participants, including those in the held-out fold. The main connection-level conditions are illustrated in Figure 3b.

The same question is also examined at the region level. This analysis uses the group-contrast cohort of 863 participants and 196 regions, with additional shrinkage fixed at $r = 0.30$. The nuisance model is first fitted to the complete analysis cohort, after which each region is ranked by the average absolute group difference across its incident connections. Region ranking is performed either with all diagnostic labels or with labels restricted to the outer training fold, allowing the effect of label access to be isolated. Models are then fitted using the top 10, 20, 30, 50, or 100 ranked regions, while a model containing all retained regions serves as the reference condition.

Pooled and within-participant symptom associations. This primary descriptive comparison examines whether out-of-fold ensemble probabilities are associated with six symptom measures. These include the ADOS total and calibrated severity scores, the ADI-R social, verbal-communication, and restricted/repetitive-behaviour domain scores, and the SRS total [11, 36, 37]. Each analysis uses only participants with an available score for the corresponding instrument, and missing symptom values are not imputed.

Associations are measured using Spearman correlation and are evaluated at two levels. The pooled analysis includes all participants with an available symptom score, whereas the within-participant analysis is restricted to autistic participants. This second analysis is included because an association observed across both diagnostic groups may partly reflect the difference between the groups rather than variation within autistic participants. Both analyses are performed using out-of-fold predictions from pooled ten-fold and leave-one-site-out validation.

3.6.2. Secondary comparison

Dynamic trajectory blocks. In this secondary comparison, the dynamic mean is evaluated both alone and together with dispersion and velocity. Two combination strategies are examined. In feature concatenation, the mean is combined with dispersion, with velocity, or with both, and a single classifier is

fitted to each joined feature vector. In late fusion, separate classifiers are fitted to the individual trajectory blocks and their predicted probabilities are averaged with equal weights. These approaches are compared because concatenation applies one regularisation penalty across all included features, whereas late fusion allows each block to be modelled separately. Standalone performance is reported for the dynamic mean, while dispersion and velocity are evaluated only as part of the combined conditions.

Classifier family. Eight classifier configurations are compared using the same static tangent features, feature scaling, and outer validation folds. This common setup ensures that performance differences are not caused by changes in the input features or participant partitions. Five linear or low-rank models are evaluated: nested ℓ_2-regularised logistic regression; ridge classification with penalty strength 100; linear support-vector classification with $C = 0.01$; shrinkage linear discriminant analysis after reduction to 200 principal components; and a rank-20 two-stage low-rank spectral classifier, which is specified below. Three tree ensembles are also included. Random forest uses 300 trees with unrestricted depth, XGBoost uses 300 trees with maximum depth 4, learning rate 0.1, row subsampling 0.8, and column subsampling 0.3, and CatBoost uses 300 trees with depth 4, learning rate 0.1, and feature subsampling 0.3. The linear and tree-based configurations are summarised in Figure 3e.

Hyperparameter tuning is applied only to the logistic model because it is the classifier used in the primary ensemble. Its C value is selected from $\{0.001, 0.01, 0.1, 1.0\}$ using three-fold stratified cross-validation within each outer training partition. All other classifiers are evaluated with fixed configurations [7, 9, 48]. The analysis is therefore treated as a secondary comparison of the specified configurations rather than a comparison of the best attainable performance of each classifier family. Before threshold-based metrics are calculated, all model outputs are placed on a common zero-to-one scale. For classifiers without `predict_proba`, the decision score is transformed with a logistic function, allowing the same decision threshold to be applied across classifiers.

Phenotype model. In this secondary comparison, a phenotype-only model is fitted using age, sex, full-scale IQ, verbal IQ, performance IQ, and handedness. Missing values are imputed using medians estimated from the outer training partition, after which the variables are standardised and entered into logistic regression. The resulting phenotype probability is evaluated both on its own and as an additional member of the imaging ensemble. This allows the contribution of phenotype information to be examined separately from the discrimination provided by the imaging models.

Atlas and fusion. This secondary comparison examines whether connectivity described by more than one brain parcellation improves discrimination. In addition to CC200, the automated anatomical labelling atlas [52] and the Harvard–Oxford atlas [18] are used, because these two atlases divide the brain on anatomical rather than functional grounds and may therefore carry complementary information. Static tangent features are computed separately for each atlas, and the probabilities of the corresponding models are averaged for every combination of two and three atlases. The same outer folds, feature scaling, and classifier are used throughout, so that any difference in performance can be attributed to the atlas information alone.

ComBat harmonisation. In this secondary comparison, ComBat harmonisation is used to reduce site-related shifts in feature means and variances [21, 31]. Site is treated as the batch variable, while age and sex are retained as covariates and diagnosis is excluded so that harmonisation does not depend on the prediction target. ComBat is fitted separately within each outer fold after feature scaling. The training features and the complete unlabelled held-out batch are supplied jointly so that a batch effect can be estimated for a site that is absent from the training data. If age or sex were missing, age would be replaced by the training median and sex would be assigned to the male category, although these rules are not required in the primary cohort. When ComBat-transformed features are unavailable, the scaled features are retained under a prespecified fallback so that every outer fold still contributes predictions.

Low-rank spectral approximation. In this secondary analysis, a linear classifier is approximated with a small number of spectral components of its weight matrix, that is, with a low-rank spectral approximation. To this end, each standardised static tangent feature vector is represented as the upper triangle of a symmetric matrix $\widetilde{S}_i$. In the same way, a dense linear coefficient vector, that is, one with a non-zero weight for every

feature, is mapped to a symmetric weight matrix W. The off-diagonal entries of W are divided by two, so that the matrix inner product matches the original vector dot product.

The weight matrix is then decomposed into its eigenvectors and eigenvalues, as given in Equation (17):

$$W = U\Lambda U^{\top}. \tag{17}$$

In Equation (17), the columns of U are the eigenvectors of W, and the diagonal matrix Λ holds the eigenvalues. Each eigenvector, together with its eigenvalue, forms one component. The components are ordered by the absolute value of their eigenvalues. This ordering places first the components that contribute most to the weight matrix. For rank q, that is, when q components are retained, the first q components then define the participant scores in Equation (18):

$$z_{ik} = u_k^{\top} \widetilde{S}_i u_k, \qquad k = 1, \ldots, q. \tag{18}$$

In Equation (18), u_k is the kth column of U, and z_{ik} is the score of participant i on component k. The scores in Equation (18) are obtained from a dense stage-one logistic model with $C = 0.01$. Next, a second logistic regression with $C = 1.0$ is fitted to the component scores. This two-stage design is used so that the final classifier operates on the q component scores alone. The ranks $\{1, 2, 5, 10, 20, 30, 50, 100\}$ are evaluated on the same folds, so that differences between ranks do not reflect different data partitions. In a separate spectral analysis, the cumulative squared-singular-value mass of the feature matrix and the cumulative absolute eigenvalue mass of W are recorded. Both masses describe how quickly the total is accumulated as components are added.

3.6.3. Exploratory comparison

Cognitive and sex subgroups. In this exploratory analysis, discrimination is examined across cognitive ability and sex using out-of-fold predictions from the primary cohort. Predictions are first matched to the correct phenotypic records after participant labels and site order are verified. Cognitive effects are then examined in three ways. First, full-scale IQ is centred within site and divided into quartiles, and AUROC is calculated within each quartile so that the groups reflect IQ relative to the site mean. Second, autistic participants are divided into full-scale IQ bands below 85, 85–99, 100–114, and 115 or above, while the same 468 non-autistic participants are used as the comparison group for every band. Third, autistic and non-autistic participants are matched on full-scale IQ within site using greedy nearest-neighbour matching without replacement and a maximum difference of five IQ points. Unmatched autistic participants are excluded, and AUROC is then estimated within site-centred IQ quartiles of the matched sample. These three cognitive-subgroup designs are illustrated in Figure 3d. Additional analyses examine the relationship between classifier scores and full-scale, verbal, and performance IQ separately within autistic and non-autistic participants. Before Spearman correlation is calculated, site, age, and mean framewise displacement are regressed from both the classifier score and IQ measure. Finally, sex-specific AUROCs are calculated only for subgroups containing at least 20 participants and both diagnostic classes, reducing instability from very small or single-class subsets.

Cohort definition. This exploratory sensitivity analysis examines whether model performance changes with cohort definition. The static tangent classifier and the primary three-member ensemble are evaluated first in the primary cohort of 871 participants and then in the wider quality-control cohort of 1,032 participants. The model specification is kept unchanged across both cohorts so that any performance difference reflects the cohort definition rather than a change in modelling choices. The static tangent classifier is evaluated with pooled ten-fold validation, while the primary ensemble is evaluated with both pooled ten-fold and leave-one-site-out validation.

Scan duration and sample size. The exploratory duration analysis examines how performance changes as the amount of scan data is reduced. Participants with at least 232 volumes are retained, and the same scans are truncated to 40, 60, 90, 120, 160, 200, and 232 volumes. This keeps the participant sample fixed across durations. Covariance matrices are re-estimated at each duration, and the analysis is repeated with additional shrinkage values of 0.15, 0.30, and 0.50 to account for the reduced number of

volumes available in shorter scans. A separate sample-size analysis examines performance using full-length scans with participant subsets of 100, 200, 300, 400, 550, 700, and 825. One unstratified random subset is used at each size, and additional shrinkage is fixed at $r = 0.15$ so that only sample size is varied. Sample size is also varied within the New York University (NYU) site using five-fold validation, allowing this effect to be examined without between-site variation. Finally, the association between scan duration or sample size and AUROC is summarised using Spearman correlation (Section 3.8.4).

Training-cohort homogeneity. In this exploratory comparison, the effect of training-cohort homogeneity on AUROC is examined. Here, homogeneity refers to whether the training participants come from one site or from several sites. For this purpose, NYU is used as the single-site source, and a static tangent model with additional shrinkage $r = 0.30$ is fitted. Four conditions are then formed by crossing one-site or multi-site training with same-site or other-site testing. In each condition, 120 training participants and 40 test participants are sampled without class stratification over 20 random repetitions. The sample sizes are held equal across conditions so that the conditions differ only in the sites that supply the participants. Across all four conditions, AUROC is used as the outcome.

Clinical-instrument concordance. This exploratory analysis examines agreement between the ADI-R and ADOS criteria within autistic participants who have complete scores for both instruments. The ADI-R criterion requires scores of at least 10 for social interaction, 8 for verbal communication, and 3 for restricted or repetitive behaviour, while the ADOS criterion requires a total score of at least 7 [36, 37]. Agreement is summarised using the four cells of the resulting 2×2 table, the discordance rate, and Cohen's κ, which measures agreement beyond chance. The ABIDE-I diagnostic label remains the classification outcome throughout. The concordance results are also combined with Social Responsiveness Scale data from non-autistic participants to examine the effect of possible label noise under the mutually contaminated model of Menon et al. [41]. The autistic-label noise rate, ε_1, is initially set to half the observed disagreement rate η, assuming that each discordant case reflects an error in one of the two instruments. The non-autistic noise rate, ε_0, is estimated from the 137 non-autistic participants with available SRS totals as the proportion whose raw score is at least 60. With π_1 and π_0 denoting the proportions of autistic and non-autistic participants in the primary cohort, the corresponding accuracy ceiling is $1 - (\pi_0\varepsilon_0 + \pi_1\varepsilon_1)$. The observed AUROC and accuracy of the primary ensemble are also adjusted using these noise rates. The corrected AUROC is $(\text{AUROC} - 0.5)/(1 - \varepsilon_0 - \varepsilon_1) + 0.5$, while corrected accuracy is $(\text{accuracy} - (\varepsilon_0 + \varepsilon_1)/2)/(1 - \varepsilon_0 - \varepsilon_1)$. The latter expression is exact for balanced accuracy but is only approximate for accuracy because the diagnostic groups are unequal in size. These noise rates are not identifiable from the available data. In particular, two diagnostic instruments applied to a single population do not uniquely determine their sensitivities, specificities, and prevalence [27], and no participant is negative on both instruments. In addition, the conventional SRS threshold of 60 applies to a T-score, whereas the release provides raw totals. For these reasons, the accuracy ceiling and corrected performance are reported as a range over $\varepsilon_1 \in \{0, \eta/2, \eta\}$ and raw-total thresholds of 40, 50, 55, 60, and 70.

Coefficient and SHAP summaries. In this exploratory analysis, the model coefficients are treated as descriptive parameters. A coefficient reflects the contribution of a feature to the prediction given all other features, rather than a group difference in that feature. For this reason, the interpretation of group differences is based on the residualised analyses in Section 3.7.

Shapley additive explanations (SHAP) are then calculated to describe how each feature contributes to individual predictions. Under SHAP, each feature is assigned a share of the difference between a participant's model output and the average output over a background set. For a linear model and a training-fold background, the SHAP values are calculated with Equation (19):

$$\phi_{ij} = w_j \left(z_{ij} - \mathbb{E}_{\mathcal{T}}[z_j] \right). \tag{19}$$

In Equation (19), ϕ_{ij} denotes the SHAP value of feature j for participant i, w_j is the corresponding model coefficient, z_{ij} is the standardised feature value, and $\mathbb{E}_{\mathcal{T}}[z_j]$ is the training-fold mean of that feature. SHAP values are computed out of fold using $C = 0.01$ under pooled ten-fold validation, with the corresponding training partition used as the explainer background [38]. This keeps held-out participants separate from the

explanation process. Within each fold, the computed SHAP values are also compared with the closed-form values using Pearson correlation to verify agreement. For interpretation, mean absolute SHAP values are first calculated for individual edges so that positive and negative contributions do not cancel. These values are then aggregated into Yeo-7 system pairs using system labels assigned from the parcel centroids. This assignment differs from the voxel-overlap mapping used in Section 3.7. Unassigned edges are omitted, while within-system edges contribute twice to diagonal cells. Finally, system-level summaries and three participant-level waterfall examples are retained to illustrate how feature contributions combine to produce individual model outputs.

Component–symptom associations. In this exploratory analysis, a rank-20 spectral classifier is fitted once to the complete primary cohort so that all participants are represented using the same set of components. The tangent reference, scaler, and classifier are therefore estimated from the full cohort and its diagnostic labels, while symptom scores are used only in the subsequent association analysis and do not influence component fitting. Within autistic participants, each component score is related to the symptom measures after age, sex, full-scale IQ, and site are regressed from both variables. Spearman correlation is then calculated between the resulting residuals, and p-values are adjusted across all component–instrument combinations. Because these associations rely on components obtained from a single full-cohort fit, component stability is also evaluated. Components estimated within each cross-validation training fold are matched to those from the full cohort, and stability is summarised by the highest absolute cosine similarity for each component. The absolute value is used because the sign of an eigenvector is arbitrary.

3.7. Exploratory group-difference analyses

3.7.1. Connection construction and nuisance adjustment

The group-difference analyses directly compare functional connectivity between autistic and non-autistic participants. These exploratory analyses use the group-contrast cohort of 863 participants and 196 retained regions. Pairwise Pearson correlations are Fisher transformed and vectorised before analysis. For each connection, a linear nuisance model is fitted using age, sex, mean framewise displacement, and acquisition-site indicators. The resulting adjusted connectivity values are then used for the group comparisons. Missing nuisance values are replaced with the median of the analysis cohort so that all participants can be retained.

Standardised group differences are then calculated from the residual connection values. For connection e, the effect is defined by Equation (20):

$$d_e = \frac{\overline{R}_{e,\text{autistic}} - \overline{R}_{e,\text{comparison}}}{\text{SD}(R_e)}. \tag{20}$$

In Equation (20), SD is the standard deviation of the residuals across both groups, while the barred terms denote the corresponding group means. The comparison group represents non-autistic participants, so negative values indicate lower connectivity in autistic participants. Standardising by the pooled residual variation places all connection effects on a common scale, allowing them to be compared and averaged.

3.7.2. Exploratory functional-system contrasts

Connection-level effects are aggregated into functional-system contrasts by grouping connections according to the systems of their two endpoints. For each system pair with at least 20 connections, the standardised connection effects are averaged to obtain a system-level effect. Statistical significance is then assessed by permuting diagnostic labels within site. The nuisance model and its residuals are fitted once before permutation, and only the group contrast is recomputed for each reshuffled label set. This keeps the nuisance adjustment fixed while testing whether the observed system-level effect is larger than expected under relabeling.

3.7.3. Exploratory motion-matched sensitivity analysis

The sensitivity of the group contrasts to head motion is examined using motion-matched samples. Autistic and non-autistic participants are matched within site by mean framewise displacement using greedy matching without replacement. Matching within site keeps acquisition site balanced between the two groups. Before matching, missing nuisance values are imputed using medians from the complete pre-matching cohort so

that imputation does not depend on the final matched sample. The group-contrast and motion-matching procedures are illustrated in Figure 3c.

Two matching calipers are used to balance motion similarity against retained sample size. For the system-level analysis, the maximum allowed difference in mean framewise displacement is 0.01 mm, producing 256 matched pairs (512 participants). For the regional analysis, a 0.02 mm caliper is used, producing 280 pairs (560 participants). Nuisance adjustment and site-stratified permutation testing are then repeated within each matched cohort. Thus, the matched analyses use the same group-contrast procedure as the full cohort but with different participant samples. The difference in mean framewise displacement between groups is also tested with the Mann–Whitney U test before and after matching to determine how effectively the matching reduces motion imbalance.

The relation between head motion and overall connectivity is examined separately. For each participant, mean connection strength is calculated as the average Fisher-transformed correlation across the 19,110 retained connections. Its association with mean framewise displacement is then estimated using Spearman correlation across the group-contrast cohort. The analysis is repeated after acquisition site is regressed from both variables. This site-adjusted estimate is included to reduce the influence of site-level differences in both motion and connectivity.

3.7.4. Exploratory regional analysis and internal stability

In this exploratory analysis, regional effects are examined within a selected functional system, with the somatomotor system used as the default. For each parcel, the regional effect is calculated as the mean standardised group difference across its within-system connections, and p-values are adjusted across the parcels tested within that system. Because this analysis is performed after inspection of the system-level results, it is treated as exploratory. The stability of the resulting regional ranking is then assessed using 200 random split-half repetitions. In each repetition, the cohort is divided into two independent halves, and nuisance adjustment and regional effects are refitted separately in each half. Agreement between the two regional rankings is measured using Spearman correlation. The same procedure is also applied to all within-system connections and to the top 50 connections. As an additional reference, 100 randomly selected region sets of the same size are evaluated using the first 50 split-half repetitions, allowing the stability of the target system to be compared with that expected from similarly sized random sets.

3.8. Statistical analysis

3.8.1. Analysis status and multiplicity

All prediction and group-difference analyses are interpreted within a common evidential framework. No confirmatory hypothesis-testing family is predefined. Instead, analyses are classified as primary descriptive, secondary, or exploratory, and this status is stated in each analysis heading. The primary descriptive results include the primary-cohort ensemble, the matched-shrinkage comparison of static and windowed connectivity, the feature-selection scope analysis, and the pooled and within-participant symptom associations. Pairwise model comparisons, including those within the primary descriptive analyses, are therefore interpreted descriptively rather than as confirmatory hypothesis tests.

Multiplicity is handled according to the type of analysis. Pairwise comparison p-values are reported without adjustment and are presented together with effect sizes. Benjamini–Hochberg q-values are also provided across these pairwise comparisons for reference, but they are not used as a decision rule. In contrast, Benjamini–Hochberg false-discovery-rate correction [6] is applied to families of related network, regional, symptom-association, and component–symptom tests. For the pooled and within-participant symptom analyses, the six symptom measures form one testing family within each combination of validation protocol and correlation scope.

3.8.2. Paired predictive comparisons

Conditions evaluated on the same participants are compared using their stored out-of-fold predictions. Differences in AUROC are tested with DeLong's test [15], while differences in binary decisions at the 0.5 threshold are assessed with McNemar's test [40]. These paired tests are used to compare discrimination and threshold-based classification under the same participant sample.

3.8.3. Permutation analyses

In the functional-system and regional analyses, 5,000 permutations of diagnostic labels are performed within site. Labels are permuted within site so that each permuted dataset keeps the number of autistic and non-autistic participants at every site. The two-sided p-value includes a plus-one adjustment and is given by Equation (21). Here T_{obs} is the observed system- or region-level effect, and T_b^* is the corresponding effect after permutation b:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\left(|T_b^*| \geq |T_{\text{obs}}|\right)}{B + 1}, \qquad B = 5000. \tag{21}$$

In Equation (21), B is the number of permutations, and $\mathbb{1}(\cdot)$ equals one when its condition holds and zero otherwise. The equation adds one to the numerator and denominator, because the observed labelling is counted as one of the possible labellings. As a result, the smallest attainable value at $B = 5000$ is $1/5001$, and values at that floor are reported as $p < 2 \times 10^{-4}$.

3.8.4. Correlation analyses

Spearman rank correlation is used in the symptom, cognitive-subgroup, motion–connectivity, internal-stability, and scaling analyses. This coefficient is used because it measures monotonic association without assuming a linear relation or normally distributed variables. In addition, each coefficient is accompanied by its participant scope, available sample size, and adjustment procedure. These details are reported because the participant set and the adjustment differ between analyses.

3.9. Reproducibility and scope

All analyses are implemented with fixed random seeds and stored outputs so that the main experiments, participant splits, and prediction results can be reproduced consistently. Fold-level results and out-of-fold predictions are retained, and participant labels and site information are checked when predictions are linked with phenotypic variables or used in paired comparisons. The scope of the findings is defined by the datasets and analysis settings used in each experiment. Primary diagnostic-group discrimination is evaluated in the 871-participant primary ABIDE-I cohort, while the wider quality-control cohort of 1,032 participants is used for cohort sensitivity analyses. Group-difference analyses use 863 participants and 196 retained regions. Generalisation is assessed across the acquisition sites represented in ABIDE-I, and the results therefore remain specific to this multisite research setting. Dynamic analyses are limited to the 50-volume, six-window design, and their temporal duration varies across sites because repetition time is not harmonised. Group-level findings are also conditional on the selected preprocessing pipeline, CC200 parcellation, nuisance adjustment, and recorded mean framewise displacement. These design choices define the conditions under which the reported results should be interpreted.

[FIXED UNTIL THIS PART - R.U. KARIM]

4. Results

The results are reported in the order of the analyses described in Section 3. First, the discrimination of the primary ensemble and its transfer to unseen sites are presented, together with a placement against published ABIDE-I work. Then, the six contributions are reported in turn. In order, these concern cognitive ability, functional-system differences under motion matching, feature-selection scope, classifier family, the scope of the symptom-severity associations, and the covariance shrinkage applied by the windowed description. Finally, the secondary results are summarised, and their null findings are stated as limits. Throughout, AUROC and accuracy are labelled as participant-pooled or as fold means (Section 3.4.2), because the two summaries can differ.

4.1. Classification performance and transfer to unseen sites

The highest AUROC of the connectivity-only configurations is obtained with the primary ensemble under both validation protocols (Table 2). Specifically, a fold-mean AUROC of 0.775 is obtained under pooled ten-fold validation, and a fold-mean AUROC of 0.773 is obtained under leave-one-site-out validation. However, the gain over the best single representation, the windowed tangent mean, is small (+0.009 in

participant-pooled AUROC). Moreover, this gain is not distinguished from chance (unadjusted DeLong $p = 0.066$), so it is not presented as an established improvement. The rank-mean ensemble (Section 3.3.4) is, in turn, reported for comparison only. Because its score is not a probability, it has no 0.5 decision threshold, and no accuracy is reported for it.

Table 2
Discrimination of the ensemble and of its members in the primary cohort (871 participants). AUROC and accuracy are fold means, and SD is the standard deviation across outer folds. The rank-mean ensemble has no probability scale, so no accuracy is given for it.

	Pooled ten-fold			Leave-one-site-out		
Configuration	AUROC	SD	Accuracy	AUROC	SD	Accuracy
Ensemble, probability mean (adopted)	0.775	0.040	0.707	0.773	0.087	0.684
Windowed tangent, mean	0.767	0.044	0.695	0.771	0.082	0.670
Ensemble, rank mean	0.756	0.038	–	0.750	0.091	–
Static tangent	0.756	0.048	0.692	0.757	0.104	0.643
Correlation	0.750	0.040	0.683	0.743	0.083	0.655

As shown by participant-pooled estimates with intervals, discrimination is maintained when a whole site is withheld from training (Table 3; Figure 4a). Specifically, an AUROC of 0.776 (95% DeLong interval 0.745–0.807) is obtained under pooled ten-fold validation. Similarly, an AUROC of 0.774 (0.744–0.805) is obtained under leave-one-site-out validation. In addition, sensitivity and specificity are nearly equal under pooled ten-fold validation (0.705 and 0.709), so the ensemble does not favour one class under this protocol. With the site withheld, in contrast, sensitivity is higher than specificity (0.742 against 0.684). As a result, more of the errors under this protocol fall on non-autistic participants. These intervals, however, condition on the fitted model and treat dependent out-of-fold predictions as independent (Section 3.4.3). For this reason, they are the narrowest defensible intervals rather than a full account of uncertainty.

Variation between sites is summarised by a random-effects model, because a centre that applies the model needs the AUROC expected at a site not yet seen. When the 20 per-site estimates are combined, therefore, a random-effects mean AUROC of 0.803 is obtained (Figure 4b). In addition, between-site heterogeneity is not distinguishable from chance ($Q = 23.5$ on 19 degrees of freedom, $p = 0.22$; $I^2 = 19\%$). As a result, the 95% prediction interval of Equation (13) for a site not yet seen extends from 0.725 to 0.882. However, leave-one-site-out estimates share training data, so the heterogeneity is, if anything, understated.

Site-level AUROC nevertheless varies widely, from 0.589 at SBL to 0.912 at UM_2. In particular, no site's point estimate falls to chance, but 2 of 20 sites lie below 0.65 (SBL and MAX_MUN). Moreover, the 95% intervals of 5 sites (CALTECH, LEUVEN_1, MAX_MUN, OHSU and SBL) include 0.5. This spread, however, is not taken as evidence of site-dependent failure, because the median site contributes about thirty participants. For this reason, wide per-site intervals are expected even under identical performance at every site.

Phenotypic variables carry some diagnostic information, but they add little to the connectivity features. Specifically, an AUROC of about 0.62 is reached by the phenotype model (Section 3.6.2) on its own. In addition, participant-pooled AUROC is raised by about 0.004 when this model is added to the ensemble (unadjusted $p = 0.041$; $q = 0.11$, for reference only). Because this gain is small, the phenotypic member is not adopted.

The placement against published ABIDE-I work is given in Table 4, in which sample size and testing procedure are listed because both change what a value means. For this comparison, the wider quality-control cohort is used, because its quality-control definition admits 1,035 participants, the same eligible count as in the published studies. However, three of these participants have no usable time series in the release, so 1,032 are analysed. In this cohort, a participant-pooled accuracy of 0.706 (95% Wilson interval 0.678–0.733) is obtained under pooled ten-fold validation. In comparison, accuracies of 0.700, 0.703 and 0.708 are reported by Heinsfeld et al. [24], Eslami et al. [20] and Almuqhim and Saeed [3], and these values are not recomputed here.

Table 3
Participant-pooled performance of the primary ensemble in the primary cohort (871 participants), with 95% intervals. AUROC carries DeLong intervals; accuracy, sensitivity and specificity carry Wilson intervals.

Validation	AUROC	Accuracy	Sensitivity	Specificity
Pooled ten-fold	0.776 (0.745–0.807)	0.707 (0.676–0.736)	0.705 (0.658–0.747)	0.709 (0.667–0.749)
Leave-one-site-out	0.774 (0.744–0.805)	0.711 (0.680–0.740)	0.742 (0.697–0.782)	0.684 (0.640–0.724)

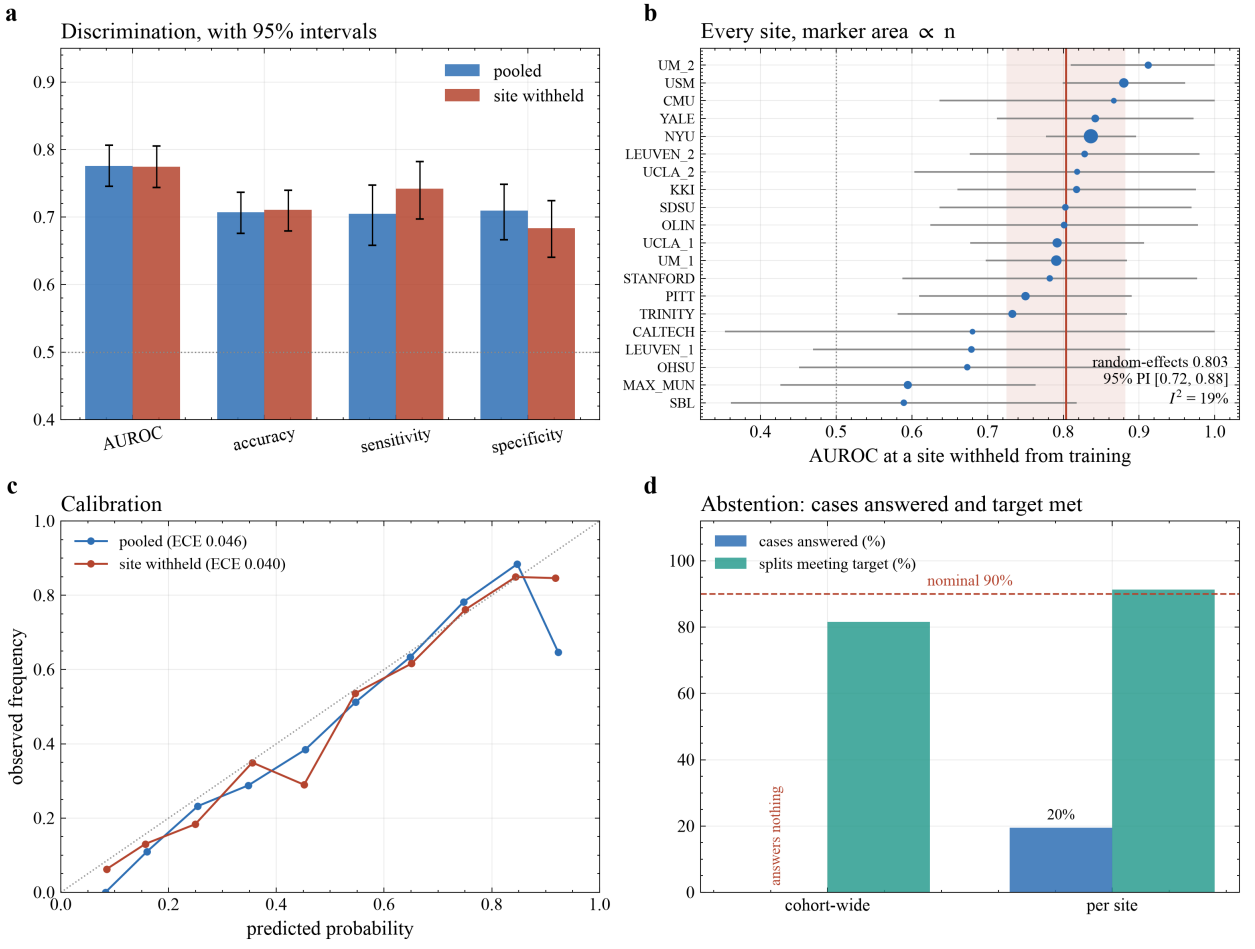


Figure 4: Discrimination, per-site transfer, calibration and abstention of the primary ensemble. (a) Discrimination with 95% intervals under both validation protocols. (b) AUROC at every site when that site is withheld from training, with marker area proportional to the number of participants; the random-effects mean and the prediction interval for a site not yet seen are shown. (c) Predicted probability against observed frequency. (d) For the cohort-wide and per-site abstention rules under leave-one-site-out validation, the percentage of participants answered when the threshold is set on the whole sample and the percentage of held-out half-splits in which the error target is met, against the nominal 90%.

The accuracy comparison in Table 4 does not separate the methods, because none of the comparison studies reports a confidence interval. In addition, the intervals implied by these sample sizes are roughly ±0.03, whereas the three published values span 0.008. For this reason, two further quantities that those studies do not report are listed, and they describe the present work without being compared with the published values. Specifically, the AUROC on an unseen site is 0.774 against 0.776 within the cohort, and the expected calibration error is 0.046 (Section 4.8). In addition, accuracy at the published level is reached

Table 4
Placement against published work on ABIDE-I resting-state functional connectivity. Values for the other studies are as reported by their authors; n.r., not reported. For the ensemble rows, AUROC is participant-pooled and accuracy is the fold mean, which agrees with the participant-pooled value to three decimals. For the tangent-model row, AUROC and accuracy are fold means. The unseen-site column gives the leave-one-site-out AUROC, with its difference from the pooled ten-fold value in brackets.

Study	Accuracy	AUROC	Sample	Procedure	AUROC, unseen site	Calibration error
Heinsfeld et al. [24]	0.700	n.r.	1,035	10-fold	n.r.	n.r.
Eslami et al. [20]	0.703	n.r.	1,035	10-fold	n.r.	n.r.
Almuqhim and Saeed [3]	0.708	n.r.	1,035	10-fold	n.r.	n.r.
This work, ensemble	0.707	0.776	871	nested 10-fold	0.774 (-0.002)	0.046
This work, ensemble, matched sample	0.706	0.789	1,032	nested 10-fold	0.783 (-0.006)	0.023
This work, tangent model, matched sample	0.696	0.766	1,032	nested 10-fold	–	–

Table 5
AUROC by full-scale IQ (FIQ) band of the autistic participants, each band scored against the same comparison group of all 468 non-autistic participants.

Autistic participants	n	Pooled ten-fold	Leave-one-site-out
FIQ below 85	40	0.873	0.871
FIQ 85–99	97	0.815	0.816
FIQ 100–114	126	0.781	0.788
FIQ 115 or above	113	0.719	0.709

under a procedure in which no step is fitted before the data are split. This property matters because a step fitted before the split can inflate the estimate (Section 4.4).

The cohort definition itself changes the score, so both definitions are reported for a like-for-like comparison. Specifically, a fold-mean AUROC of 0.756 is reached by the static tangent model in the primary cohort of 871 participants, and 0.766 in the wider cohort of 1,032 participants. However, the scans excluded by the primary definition are mostly those with head motion or failed processing, and they are not evenly split between the groups. Part of this gain may, therefore, reflect data quality rather than differences in brain connectivity.

4.2. Cognitive ability and detection

Discrimination declines as the cognitive ability of the autistic participants rises, and this pattern underlies the first contribution of the study. Because the subgroup analyses are exploratory (Sections 3.6.3 and 3.8.1), the pattern is reported as a description rather than as a confirmed effect. To separate it from changes in the comparison group, the comparison group is held fixed at all 468 non-autistic participants. Only the autistic participants, in turn, are divided into full-scale IQ bands. As a result, every band is scored against the same reference, so differences between bands cannot arise from changes in the comparison group. Under this design, AUROC falls monotonically from 0.873 below full-scale IQ 85 to 0.719 at 115 or above under pooled ten-fold validation (Table 5; Figure 5d). Similarly, AUROC falls from 0.871 to 0.709 when the site is withheld. However, the autistic bands still differ from the reference group in full-scale IQ, and this separation is addressed next.

The gradient is reproduced when both groups are stratified by IQ, but this design is also affected by the group difference in IQ. Specifically, all participants are divided into quartiles of site-centred full-scale IQ (Table 6). Under pooled ten-fold validation, AUROC then falls from 0.833 in the lowest quartile to 0.709 in the highest. However, autistic participants have lower full-scale IQ (Table 1), so the lower quartiles are also those in which the groups differ most on IQ. For this reason, each autistic participant is matched to a

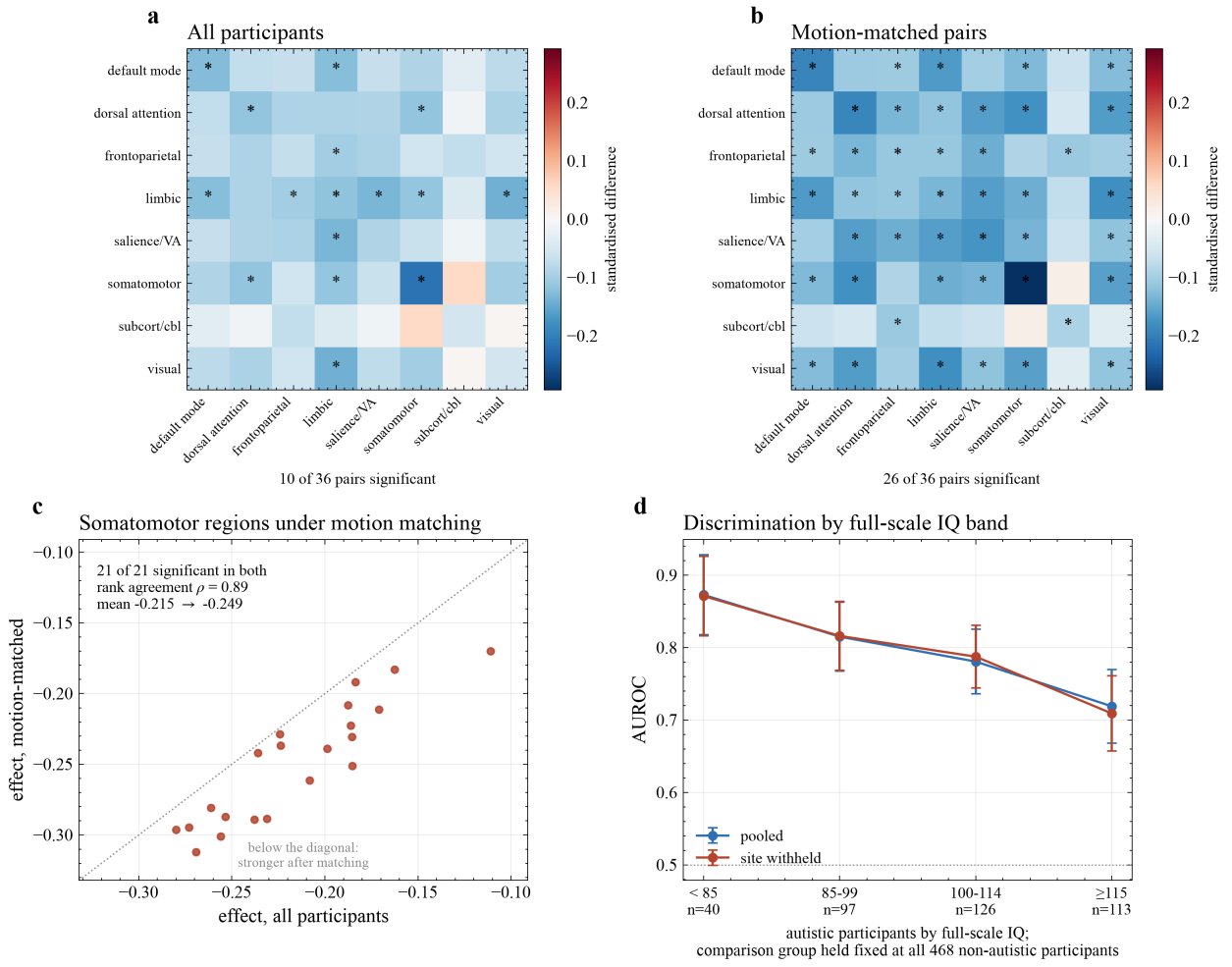


Figure 5: Group differences in functional-system connectivity and the effect of motion matching. (a, b) Standardised difference for every pair of systems in all participants and in within-site pairs matched for head motion, on a shared colour scale; asterisks mark pairs with Benjamini–Hochberg $q < 0.05$. (c) The effect of each somatomotor region in all participants plotted against its effect in motion-matched pairs; points below the diagonal are stronger once motion is matched. (d) Discrimination by full-scale IQ band among autistic participants (below 85, 85–99, 100–114, and 115 or above), each against all 468 non-autistic participants, under both validation protocols.

non-autistic participant at the same site whose full-scale IQ differs by at most five points. As a result, 274 pairs are obtained under pooled ten-fold validation and 275 pairs with the site withheld.

After IQ matching, discrimination remains highest in the lowest quartile but no longer declines across the upper three. Specifically, AUROC in the lowest quartile is 0.855 under pooled ten-fold validation and 0.848 with the site withheld. In contrast, AUROC across the upper three quartiles ranges from 0.700 to 0.756 under pooled ten-fold validation and from 0.735 to 0.756 with the site withheld. Therefore, part of the gradient over the full IQ range reflects the group difference in IQ. The within-group association, which involves no comparison between groups, is for this reason the direct evidence for the effect.

Within the autistic group, classifier scores decline with cognitive ability, whereas no such association is found in non-autistic participants (Table 7). For this analysis, site, age and mean framewise displacement are regressed from both the score and the IQ measure (Section 3.8.4). Under this adjustment, the score correlates with full-scale IQ at $\rho = -0.180$ ($p = 0.0004$, $n = 376$) among autistic participants under pooled ten-fold validation. In contrast, the corresponding value among non-autistic participants is +0.048

Table 6
AUROC by quartile of site-centred full-scale IQ, in all participants and in participants matched on full-scale IQ within site. The matched counts are given for pooled ten-fold and leave-one-site-out (LOSO) validation, respectively.

	All participants			IQ-matched participants		
Quartile	n	Pooled ten-fold	LOSO	n (pooled / LOSO)	Pooled ten-fold	LOSO
Q1	204	0.833	0.825	137 / 144	0.855	0.848
Q2	203	0.794	0.793	137 / 132	0.700	0.735
Q3	201	0.746	0.745	138 / 137	0.751	0.737
Q4	203	0.709	0.705	136 / 137	0.756	0.756

Table 7
Spearman correlation between classifier score and IQ within each diagnostic group, given as ρ (p). Both variables are residualised on site, age and mean framewise displacement before correlation. The p-values are not adjusted for multiplicity.

		Autistic		Non-autistic	
Measure	n (autistic / non-autistic)	Pooled ten-fold	LOSO	Pooled ten-fold	LOSO
Full-scale IQ	376 / 435	−0.180 (0.0004)	−0.179 (0.0005)	+0.048 (0.3208)	+0.022 (0.6466)
Verbal IQ	342 / 379	−0.182 (0.0007)	−0.189 (0.0004)	+0.039 (0.4547)	+0.001 (0.9892)
Performance IQ	344 / 391	−0.152 (0.0047)	−0.147 (0.0062)	+0.007 (0.8862)	−0.005 (0.9271)

($p = 0.3208$, $n = 435$). In addition, the same pattern is obtained for verbal and performance IQ and under both validation protocols.

The absence of an association in non-autistic participants constrains the interpretation of this result. Because no gradient is found in that group, the association is not attributed to a general relation between ability and classifier score. However, the two correlations are not compared by a direct test, and full-scale IQ varies less among non-autistic participants (Table 1). For this reason, this interpretation remains provisional. In addition, Nielsen et al. [43] report that classifier scores covary with verbal IQ in ABIDE data, so the association is anticipated. Beyond that report, the comparison with non-autistic participants and the graded loss of detectability across IQ bands are added by the present analysis.

Taken together, these results indicate that measured performance on ABIDE-I depends on the cognitive composition of the autistic sample. In addition, connectivity-based detection is least reliable in cognitively able autistic individuals. This dependence matters because a cohort with a different distribution of cognitive ability can yield a different AUROC from the same method.

No deficit in discrimination is found in female participants, although this subgroup is small. Specifically, AUROC is 0.827 (0.760–0.895, $n = 144$) in female participants and 0.764 (0.730–0.798, $n = 727$) in male participants under pooled ten-fold validation. Similarly, with the site withheld, AUROC is 0.806 (0.735–0.878) in female participants and 0.767 (0.733–0.801) in male participants. However, the intervals overlap, so no difference between the sexes is claimed.

4.3. Functional-system differences under motion matching

Group differences are assessed in exploratory analyses (Sections 3.7 and 3.8.1), and the strengthening of the somatomotor reduction under motion matching is the second contribution of the study. These differences are compared at the level of functional systems, because the 19,110 individual connections are too many to test one at a time with useful power. For this purpose, the regions are grouped into the seven cortical systems of Yeo et al. [57] and a subcortical and cerebellar class (Section 3.1.5). As a result, the comparisons are reduced to 36 system pairs, each averaging over 21 to 1,462 connections (median 441). In addition, every connection is first adjusted for acquisition site, head motion, age and sex. Significance is then obtained by permuting diagnosis within each site (Sections 3.7.1, 3.7.2 and 3.8.3).

In the 863 participants of the group-contrast cohort, 10 of 36 system pairs differ after correction, and every one of them is lower in autism (Table 8; Figure 5a). Moreover, 34 of 36 pairs are negative overall, and no pair is significantly raised. Among these pairs, the largest difference is found within the somatomotor system

Table 8
Functional-system pairs with the largest standardised group differences in the group-contrast cohort (863 participants). Negative values indicate lower connectivity in autistic participants, and q is the Benjamini–Hochberg-adjusted permutation p-value. The first ten rows are the pairs with $q < 0.05$.

System	System	Connections	Difference	z	q
Somatomotor	Somatomotor	210	−0.215	−4.769	0.002
Limbic	Visual	210	−0.140	−3.633	0.002
Limbic	Salience / ventral attention	147	−0.133	−3.247	0.013
Default mode	Default mode	903	−0.127	−3.777	0.002
Default mode	Limbic	301	−0.125	−3.677	0.004
Limbic	Limbic	21	−0.118	−3.052	0.014
Dorsal attention	Somatomotor	399	−0.116	−2.653	0.024
Dorsal attention	Dorsal attention	171	−0.115	−2.875	0.024
Limbic	Somatomotor	147	−0.114	−2.839	0.024
Frontoparietal control	Limbic	147	−0.102	−2.733	0.024
Somatomotor	Visual	630	−0.101	−2.317	0.058
Frontoparietal control	Salience / ventral attention	441	−0.093	−2.317	0.058

(standardised difference −0.215, $z = -4.769$, $q = 0.002$). In addition, the limbic system is the most widely involved and differs from most other systems. Because site enters both the adjustment and the permutation, differences in mean connectivity between acquisition sites are removed before this reduction is estimated.

The permutation certifies the p-value on residualised connections, but it does not certify the residualisation itself. For this reason, within-site pairs matched for mean framewise displacement are also analysed (Section 3.7.3). This matching is needed because head motion alters measured connectivity [46], and autistic participants move more than non-autistic participants (Table 1). Under the system-level caliper, 256 pairs are retained, and the group difference in motion is removed ($p = 7.6 \times 10^{-7}$ before matching and $p = 0.59$ after). In these matched pairs, 26 of 36 system pairs are significant, against 10 of 36 in the full group-contrast cohort (Figure 5b). However, the matched sample also differs from the full cohort in size and composition, so the rise is not attributed to motion matching alone. In addition, the somatomotor within-system difference strengthens from −0.2155 ($z = -4.77$) in all participants to −0.2925 ($z = -4.89$) in the matched pairs. Age, however, is not matched and is handled by adjustment alone, so the matched analysis addresses head motion rather than age.

Inside the somatomotor system, the effect is spread across its 21 regions rather than concentrated in one (Supplementary Material). Specifically, all 21 regions differ significantly in all participants, and the strongest effects lie in the postcentral gyrus, the paracentral lobule and the precentral gyrus. In anatomical terms, these regions form primary somatosensory and motor cortex, and they are affected on both sides to a similar degree. However, this regional analysis is exploratory, because the system is chosen for carrying the largest system-level difference (Section 3.7.4). Its ordering is, therefore, a description of that result rather than an independent test of it.

The regional result also survives motion matching and becomes stronger under it. Specifically, 280 pairs (560 participants) are retained by the wider regional caliper, and the motion difference is again removed ($p = 0.28$ after matching). Under this matching, all 21 regions remain significant, every one of them strengthens, and their ordering is preserved (rank correlation 0.89 against the unmatched result). Moreover, the mean regional effect moves from −0.215 in all 863 participants to −0.249 in the matched pairs (Figure 5c).

The regional ordering is moderately reproducible within the cohort, and this reproducibility indicates how far the ranking of regions can be relied on. Specifically, a correlation of 0.55 across the 21 regions is obtained in repeated split-half analyses, with the nuisance adjustment refitted in each half. In comparison, the ordering of the 210 connections inside the system reproduces at 0.38, and that of its top fifty at 0.34. In addition, 100 random blocks of 21 regions drawn from the whole brain give a median of 0.35 (range 0.06–0.63), and 9% of them reach the somatomotor value. The ordering is, therefore, more stable than most blocks of equal size, but not categorically so.

Table 9
Fold-mean AUROC under the seven data-access conditions, with 19,900 static tangent features in the primary cohort (871 participants) and identical outer folds. SD is the standard deviation across outer folds.

Condition	AUROC	SD
Reference: every step fitted within the training partition	0.756	0.048
Tangent reference fitted on all participants	0.756	0.048
Tangent reference and scaler fitted on all participants	0.753	0.049
Regularisation constant chosen on the held-out fold	0.758	0.048
Acquisition site added as a predictor	0.756	0.049
Tangent reference fitted on all participants and 2,000 of 19,900 connections selected with all diagnostic labels	0.987	0.008
All five changes combined	0.988	0.008

The reduction runs against the direction expected from residual motion. Specifically, mean framewise displacement correlates positively with mean connection strength ($\rho = +0.51$ across participants and $+0.45$ within site). Higher motion is, therefore, associated with higher rather than lower connectivity in this pipeline. Because autistic participants move more, residual motion would on average be expected to raise their connectivity, which is opposite to the observed reduction. In addition, the somatomotor location of the difference agrees with the somatomotor involvement reported by Chen et al. [8]. Its direction also agrees with the somatomotor reductions reported by Holiga et al. [25] and Ilioska et al. [29] in samples that include ABIDE data. Because those samples may share participants with the present cohort, this agreement is not an independent replication. The present result, in contrast, is obtained in within-site pairs matched for head motion.

These group differences carry three limits. First, residual motion is not a sufficient explanation for them, but it is not thereby excluded as a contributor. Second, the contrasts are univariate and are computed without reference to the classifier, so they support no statement about which measurements the classifier uses. Finally, all group differences are obtained under a single denoising strategy without global signal regression, so their direction is conditional on that choice.

4.4. Feature-selection scope and granularity

The effect of data-access scope on measured performance is examined in seven conditions. Among these conditions, the inflation produced by feature selection outside the training partition, and its dependence on feature granularity, form the third contribution. For this purpose, the conditions are compared on identical folds with the 19,900 static tangent features of the primary cohort (Section 3.6.1; Table 9). Under this design, fold-mean AUROC is changed by at most 0.003 when the tangent reference, or the reference and the scaler, is fitted on all participants. Similarly, it is changed by a similar amount when the regularisation constant is chosen on the held-out fold (0.758) or when acquisition site is added as a predictor (0.756).

In contrast, fold-mean AUROC is raised from 0.756 to 0.987 when 2,000 of 19,900 connections are selected with an F test on all diagnostic labels before splitting. Likewise, accuracy rises from 0.692 to 0.948 on the same folds. Because fitting the tangent reference on all participants leaves AUROC at 0.756 on its own, this rise is attributed to the selection step. As a result, an inflation of 0.231 in AUROC is produced by this step, and this inflation is the largest effect in the study. Moreover, 0.988 is obtained when every change is applied together, so the remaining changes add little beyond the selection step. Therefore, of the changes examined, only feature selection with all labels changes the score by more than 0.003. By comparison, choosing the regularisation constant on the held-out fold also uses held-out labels, but its effect lies within 0.003. As a consequence, figures from pipelines that select connections before cross-validation are not comparable with estimates in which selection is nested.

Selection at the level of whole regions, in contrast, gives no consistent advantage (Table 10; Figure 6c). For this comparison, regions are ranked either with all diagnostic labels or with the labels of each training fold only (Section 3.6.1). Under this design, the two scopes differ by 0.006 on average in absolute value across 10 to 100 of 196 regions. Specifically, the difference is 0.001 at 10 regions and at most 0.012 at 100 regions, and its sign is inconsistent. Therefore, the risk of a selection step depends on feature granularity. In particular,

Table 10
Participant-pooled AUROC when the top-ranked regions are chosen with all diagnostic labels or with the labels of each training fold only, in the group-contrast cohort (863 participants and 196 regions). All 196 regions give 0.781.

Regions kept	Chosen with all labels	Chosen within each fold
10	0.628	0.627
20	0.660	0.656
30	0.699	0.702
50	0.738	0.748
100	0.764	0.752

Table 11
Classifier family on the static tangent features of the primary cohort (871 participants), with identical features, scaling and outer folds. Only logistic regression is tuned; the other classifiers use fixed configurations.

Classifier	AUROC (participant-pooled)	AUROC (fold mean)	SD (folds)	Accuracy (fold mean)
Logistic regression	0.757	0.756	0.048	0.692
Ridge	0.755	0.755	0.047	0.693
Linear support-vector	0.753	0.753	0.048	0.687
Shrinkage LDA (PCA-200)	0.741	0.739	0.052	0.669
Low-rank spectral (rank 20)	0.735	0.737	0.044	0.672
CatBoost	0.624	0.623	0.042	0.597
XGBoost	0.617	0.616	0.050	0.584
Random forest	0.611	0.609	0.049	0.577

a choice of 10 of 196 regions is too coarse to fit noise, whereas a choice of 2,000 of 19,900 connections is not. However, the two comparisons also differ in cohort (871 against 863 participants), additional shrinkage, ranking statistic and AUROC summary (Section 3.6.1). For this reason, the contrast is consistent with an effect of granularity, but granularity is not the only difference between the two comparisons.

Restricting the model to the most discriminating regions also lowers performance at every list length. Specifically, all 196 regions give an AUROC of 0.781, which is higher than any reduced set in Table 10. In particular, the smallest set loses 0.154 against the use of every region. This pattern, in turn, suggests that the information used for discrimination is spread across many regions rather than confined to those with the largest group differences. Consequently, a reduced set of regions does not improve discrimination on these data, and this restriction is reported as a limit.

Without any selection before the split, the published accuracy range is already reached by the ensemble. Specifically, a participant-pooled accuracy of 0.706 is obtained in the wider cohort under pooled ten-fold validation (Section 4.1). This value is comparable to the 0.700, 0.703 and 0.708 reported for the same eligible count of 1,035 participants. Accuracy at that level, therefore, does not require selection outside the training partition.

4.5. Classifier family

Linear classifiers outperform the fixed-configuration tree ensembles on static tangent connectivity, and this result is the fourth contribution. For this comparison, all eight classifiers use the same features, feature scaling and outer folds (Section 3.6.2), so differences are not confounded by the input or the partition. Under this design, participant-pooled AUROCs of 0.735–0.757 are reached by the five linear classifiers, whereas 0.611–0.624 is reached by random forest, XGBoost and CatBoost (Table 11). In particular, 0.757 is reached by penalised logistic regression, and ridge (0.755) and linear support-vector classification (0.753) lie close to it. In contrast, shrinkage linear discriminant analysis (0.741) and the rank-20 spectral model (0.735) are lower.

The gap between logistic regression and each tree ensemble is large relative to its uncertainty. Specifically, the participant-pooled AUROC of logistic regression is higher than that of XGBoost by 0.1396, of random forest by 0.1461 and of CatBoost by 0.1327. For these differences, unadjusted DeLong p-values range from

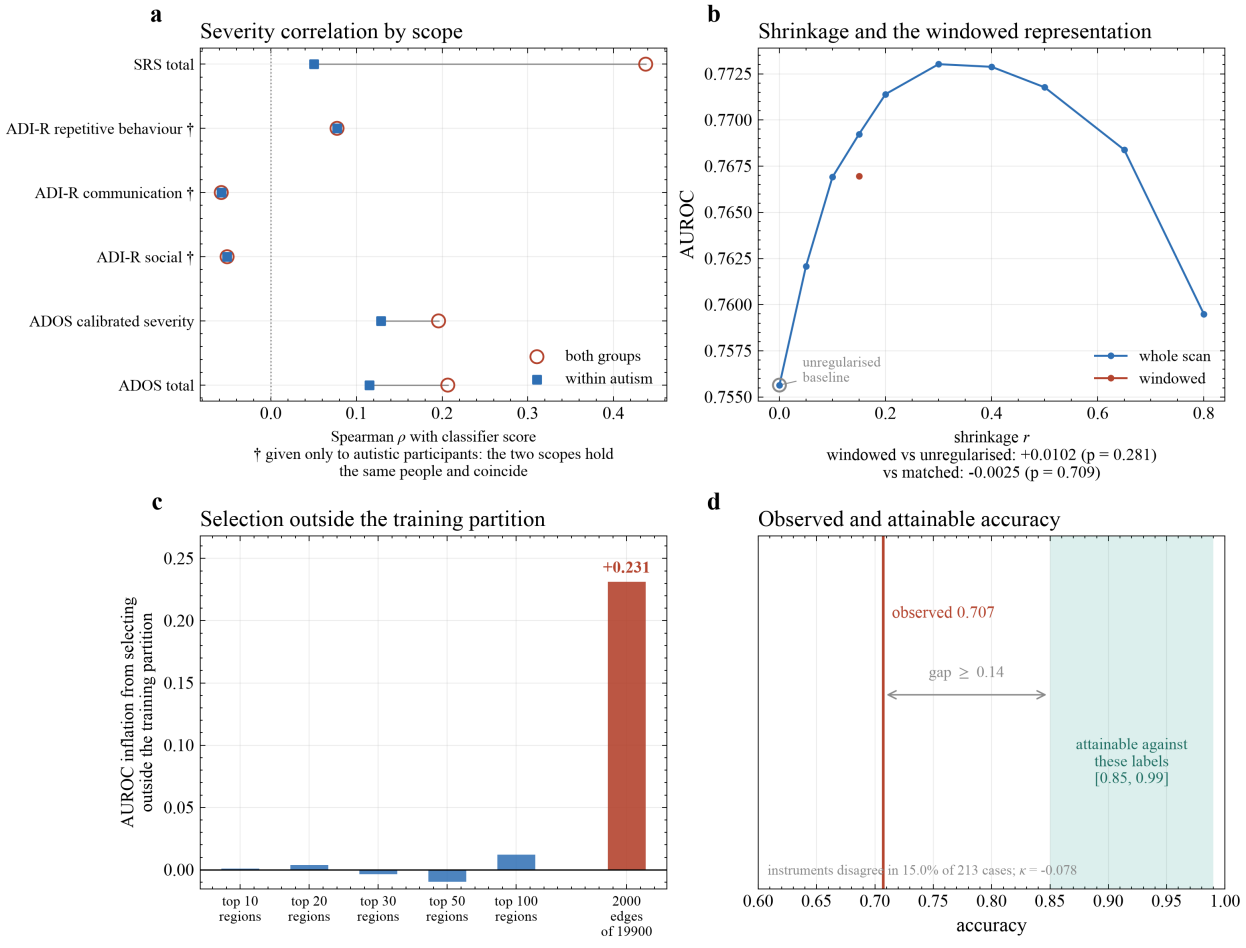


Figure 6: Four effects that change when the corresponding control is applied. (a) Correlation between classifier score and symptom severity under pooled ten-fold validation, computed across both groups and within the autistic group; rows marked with a dagger are instruments given only to autistic participants, for which the two scopes must coincide. (b) AUROC of whole-scan connectivity across shrinkage levels, with the windowed representation placed at the shrinkage that it applies to its own windows. (c) AUROC inflation from selection outside the training partition, for 10 to 100 of 196 regions and for 2,000 of 19,900 connections. (d) Observed accuracy against the interval of accuracy attainable given the diagnostic labels.

6.3×10^{-15} to 2.4×10^{-12} (Section 3.8.2). By comparison, the rank-20 spectral model lies 0.0214 below logistic regression (unadjusted $p = 0.0439$; $q = 0.1065$, for reference only).

This contrast characterises the configurations evaluated rather than the best attainable member of each family. Specifically, only the logistic model is tuned, whereas the tree ensembles use fixed configurations without nested tuning (Section 3.6.2). However, the fixed-configuration ridge, linear support-vector and discriminant classifiers also exceed all three tree ensembles. Therefore, the advantage of these linear configurations over these tree-ensemble configurations does not rest on the tuning of the logistic model alone.

4.6. Scope of the symptom-severity associations

Under pooled ten-fold validation, no association between the classifier score and symptom severity survives correction within the autistic group, and this absence forms the fifth contribution of the study (Figure 6a). For this question, the correlation is computed in two scopes, namely across both groups and within the autistic group only. The second scope is needed because non-autistic participants who

complete these instruments score at the low end of the scale. In this cohort, for example, 19 non-autistic participants have an ADOS total and 137 an SRS total. For this reason, any model that separates the groups produces a correlation across both groups largely by construction. Consistent with this, 3 of 6 associations survive Benjamini–Hochberg correction across both groups, whereas 0 of 6 do within the autistic group (Section 3.6.1).

The largest association across both groups almost vanishes once non-autistic participants are removed. Specifically, for the SRS total, a correlation with the classifier score of $\rho = 0.438$ is obtained across both groups ($n = 300$), but of 0.050 within the autistic group ($n = 163$). Similarly, values of 0.207 against 0.115 are obtained for the ADOS total, and of 0.196 against 0.128 for the ADOS calibrated severity. In contrast, identical correlations in both scopes are obtained for the ADI-R domains, which are given only to autistic participants, because both scopes contain the same people. Therefore, the correlations across both groups reflect the separation of the groups. An association with the degree of autism within the autistic group, in turn, is not established by these data.

4.7. Windowed connectivity and covariance shrinkage

The small advantage of the windowed representation is not distinguished from chance, and it is not retained when the whole-scan estimate receives the same covariance shrinkage; this result is the sixth contribution (Figure 6b). Specifically, an extra shrinkage of 0.15 is applied by the windowed representation to its window covariances, and this shrinkage is not received by the whole-scan representation (Section 3.6.1). For this reason, the same shrinkage is applied to the whole-scan estimate, with nothing else changed. Under this control, the static AUROC is raised by 0.013 through shrinkage alone (unadjusted DeLong $p = 0.004$). In contrast, the windowed representation differs from the unshrunk static baseline by 0.010 ($p = 0.281$) and from the matched static baseline by -0.002 ($p = 0.709$). Therefore, no evidence is obtained that windowing adds information beyond the shrinkage that it applies, and the apparent benefit of windowing is attributed to that shrinkage.

4.8. Secondary results

The remaining analyses are reported as descriptive results, and none of them is claimed as a contribution. In the first of these, the ensemble probabilities are found to be well calibrated under both protocols (Figure 4c). Specifically, the expected calibration error of Equation (14) is 0.046 under pooled ten-fold validation and 0.040 under leave-one-site-out validation. In addition, these values are computed from participant-pooled predictions, for which accuracy is 0.707 and 0.711, respectively. Because this error is the bin-weighted mean gap between predicted probability and observed frequency, small values indicate agreement on average. However, a small error does not guarantee agreement within every probability bin, so the reliability diagram is also shown.

Calibration, however, does not translate into a useful guarantee on the cases that the model chooses to answer (Figure 4d; Figure 7). To test such a guarantee, the leave-one-site-out predictions are divided at random into two halves, and the threshold is chosen on one half and applied to the other (Section 3.5.3). This division is needed because a threshold scored on the participants that set it passes its bound by construction. When the cohort-wide threshold is set on the whole sample, however, no participant is answered. In addition, under the half-split test, the target is met in a proportion of 0.8158 of the 38 half-splits with a finite selective risk, which is below the nominal 0.90.

The target is held by the per-site threshold, but few participants are answered by it. Specifically, when it is set on the whole sample, 170 of 871 participants (19.5%) are answered, and 143 of them are at a single site. Moreover, no participant is answered at 17 of the 20 sites. Across held-out half-splits, in addition, its mean coverage, the proportion of participants answered, is 13.7%, and the target is met in a proportion of 0.9137 of 197 valid splits. Abstention is, therefore, reported as a measurement on these data rather than as a feature of the model.

Describing connectivity with more than one atlas does not improve discrimination either. Specifically, a fold-mean AUROC of 0.756 is reached by CC200 alone, against 0.749 for CC200 with the anatomical labelling atlas, 0.755 for CC200 with the Harvard–Oxford atlas, and 0.752 for all three together (Section 3.6.2). In addition, each anatomical atlas reaches less on its own, namely 0.692 and 0.717. The functional parcellation therefore carries the information that the combinations provide, and no gain is obtained from adding atlases.

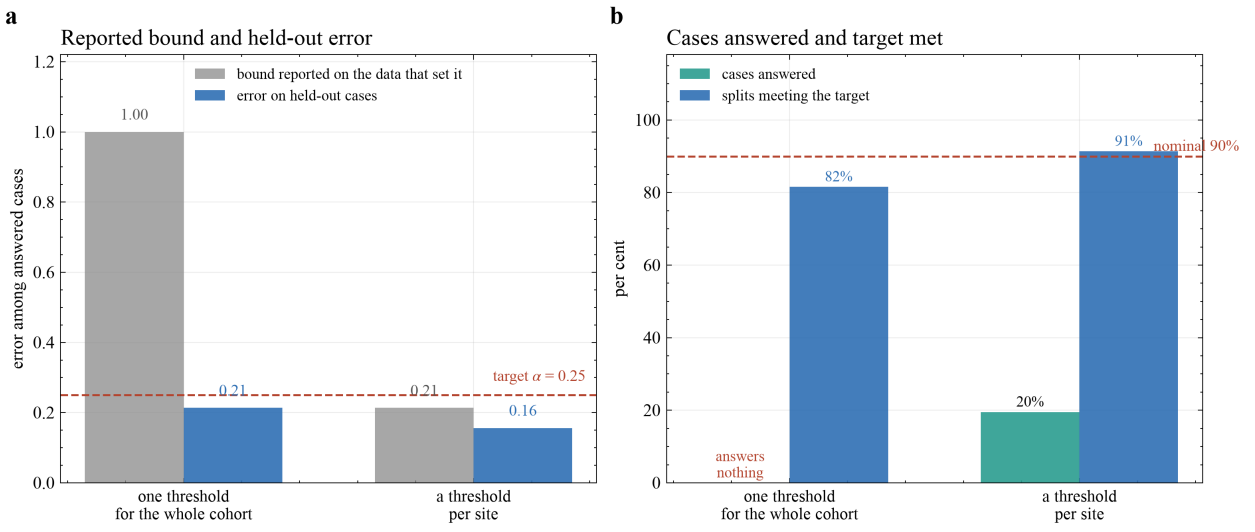


Figure 7: The error guarantee of the abstention rules under leave-one-site-out validation and its evaluation on held-out participants. (a) The error bound that each rule reports on the participants used to set it, against the error that the same rules incur on held-out cases. (b) The cost of each rule for obtaining that bound: the proportion of cases it answers, and the proportion of held-out splits in which it meets its target.

The diagnostic labels themselves limit attainable accuracy, but the size of this limit cannot be fixed from these data (Figure 6d). To estimate it, the two diagnostic instruments are compared in the 213 autistic participants with both a complete ADI-R algorithm and an ADOS total (Section 3.6.3). Under this comparison, the instruments disagree in 15.0% of these participants. Specifically, 13 meet criteria on the interview only, 19 on the observation only and 181 on both, whereas no participant is negative on both instruments. Under the assumptions of Section 3.6.3, the accuracy attainable against these labels therefore lies between approximately 0.85 and 0.99. Observed accuracy, in turn, lies 0.14 below even the lower end of that range.

At the central noise rates, correction for label noise does not close this gap. These central rates are the values defined in Section 3.6.3, namely half the observed disagreement and the proportion of non-autistic participants with an SRS raw total of at least 60. Specifically, when the correction of Menon et al. [41] is applied at these rates, accuracy is moved from 0.707 to 0.729, and AUROC from 0.776 to 0.806. The shortfall is, therefore, not explained by uncertainty in the diagnosis alone. However, at the most pessimistic rates the correction is larger, so the size of the remaining gap depends on assumptions that these data cannot settle. In particular, agreement beyond chance is slightly negative (Cohen's $\kappa = -0.078$), which is inconsistent with the two-rater model that the correction assumes. Moreover, two instruments applied to one population do not identify their error rates [27], and, as noted above, no participant is negative on both. For these reasons, no point estimate of the ceiling is reported.

5. Discussion

This study examines what connectivity-based classification of autism measures on ABIDE-I, and which analysis choices determine its reported performance. To this end, a single pipeline is used in which every learned quantity is fitted within the training partition. Under this protocol, six findings are obtained. First, detection is weakest in cognitively able autistic individuals. Second, somatomotor hypoconnectivity in autism strengthens, rather than weakens, when head motion is matched within site. Third, feature selection before the data are split inflates discrimination at the level of connections but not of regions. Fourth, linear classifiers outperform fixed-configuration tree ensembles on identical tangent-space features. Fifth, classifier scores are not associated with the degree of autism within the autistic group. Sixth, the advantage of windowed connectivity is explained by the covariance shrinkage that this description applies.

These findings matter for the interpretation of connectivity-based autism research. Because detection depends on cognitive ability, an accuracy reported on ABIDE-I partly reflects the cognitive composition of the autistic sample. Benchmarks obtained on differently composed samples are, therefore, not directly comparable. In addition, the somatomotor difference is not explained by the recorded head motion, because it strengthens once motion is matched. This robustness strengthens its standing as a candidate marker of autism. Moreover, the feature-selection result shows that reported accuracies must be read together with the procedure that produced them. In addition, simple linear models remain a strong default for high-dimensional connectivity features. The symptom result, in turn, shows that a score which separates the diagnostic groups does not thereby measure the degree of autism. Finally, the shrinkage control shows that an apparent gain from temporal information can be produced by regularisation alone.

5.1. Interpretation of the results

Discrimination is maintained when a whole acquisition site is withheld from training. The primary ensemble reaches a participant-pooled AUROC of 0.776 under pooled ten-fold validation and 0.774 under leave-one-site-out validation (Table 3). Because the two values are almost identical, the model does not rely on knowledge of the acquisition site. In addition, a random-effects mean AUROC of 0.803 is obtained across the 20 sites. The 95% prediction interval for a site not yet seen extends from 0.725 to 0.882. Such an interval is more informative for a centre that applies the model than a pooled estimate alone. However, site-level AUROC ranges from 0.589 to 0.912, and the intervals of five sites include 0.5. This spread is expected for sites of about thirty participants, so it is not read as site-dependent failure.

Accuracy at the published level is reached without any step fitted before the data are split. Specifically, a participant-pooled accuracy of 0.706 (95% Wilson interval 0.678–0.733) is obtained in the 1,032-participant cohort. In comparison, accuracies of 0.700, 0.703, and 0.708 are reported by Heinsfeld et al. [24], Eslami et al. [20], and Almuqhim and Saeed [3]. Because these values lie within one interval of each other, the comparison indicates parity rather than superiority.

The dependence of detection on cognitive ability is the first finding that concerns autism itself. Against one fixed group of 468 non-autistic participants, AUROC falls from 0.873 below full-scale IQ 85 to 0.719 at 115 or above. Within the autistic group, classifier scores also decline with full-scale IQ ($\rho = -0.18$, $p = 0.0004$). In contrast, no such association is found within the non-autistic group ($\rho = 0.05$, $p = 0.32$). The within-group association is the more direct evidence, because it involves no comparison between the diagnostic groups. When the groups are matched on IQ, however, discrimination remains highest in the lowest quartile but no longer declines across the upper three. Part of the band gradient, therefore, reflects the IQ difference between the groups. Nevertheless, an association between classifier scores and verbal IQ has also been reported in ABIDE data [43]. As a practical consequence, a connectivity-based aid would be least reliable for cognitively able autistic individuals.

The somatomotor finding shows why motion control changes the interpretation of group differences. Head motion is known to alter measured functional connectivity [46]. In this pipeline, mean framewise displacement is positively associated with mean connection strength ($\rho = 0.51$ across participants and 0.45 within site). Because autistic participants move more, residual motion would, on average, raise their connectivity, which is opposite to the observed reduction. Consistent with this reasoning, the number of system pairs with reduced connectivity rises from 10 to 26 of 36 in 256 within-site pairs matched for motion. Similarly, the mean effect of the 21 somatomotor regions strengthens from -0.215 in all 863 participants to -0.249 in 280 matched pairs. Moreover, all 21 regions remain significant under this matching, and every one of them strengthens. The direction agrees with the somatomotor involvement reported by Chen et al. [8] and with the reductions reported by Holiga et al. [25] and Ilioska et al. [29]. These studies, however, drew on samples that include ABIDE data, so the agreement is not an independent replication.

The feature-selection result shows how reported accuracies can differ widely under apparently similar pipelines. When 2,000 of 19,900 connections are selected with all labels before the split, fold-mean AUROC is raised from 0.756 to 0.987 on identical folds. In addition, accuracy is raised from 0.692 to 0.948. In contrast, selecting 10 to 100 of 196 regions with all labels changes AUROC by only about 0.006 on average. The inflation, therefore, depends on the granularity of the features. This dependence is plausible, because selection among many candidate connections can exploit chance associations with the labels of held-out participants. The two comparisons, however, also differ in cohort, shrinkage, and ranking statistic, so they

contrast two levels rather than establish a trend. For this reason, a study that selects connections before cross-validation can report inflated estimates, and its procedure must be stated together with its accuracy.

The classifier comparison indicates that linear models suit high-dimensional connectivity features. On identical features and folds, the five linear classifiers reach participant-pooled AUROCs of 0.735–0.757. In contrast, the three tree ensembles reach 0.611–0.624 (unadjusted DeLong $p \leq 2.4 \times 10^{-12}$ for the logistic model against each ensemble). This pattern is consistent with the benchmark of Dadi et al. [14], in which tangent-space connectivity performs well with regularised linear classifiers. However, the tree ensembles use fixed configurations, so the gap characterises these configurations rather than the best attainable tree model.

The symptom analysis shows what the classifier score does not measure. Across both diagnostic groups, 3 of 6 associations between the score and symptom severity survive correction, whereas 0 of 6 survive within the autistic group. For the SRS total, for example, the correlation is 0.438 across both groups but 0.050 within the autistic group. This contrast is expected, because few non-autistic participants complete these instruments, and those who do score at the low end of the scale. Any model that separates the groups therefore produces a correlation across both groups largely by construction. The score is, for this reason, read as a group-separating measurement rather than as an index of the degree of autism. This distinction matters, because such correlations are often presented as evidence that a model captures a graded phenotype.

The shrinkage control shows that an apparent advantage of windowed connectivity need not arise from temporal information. An additional covariance shrinkage of 0.15 is applied by the windowed description to its window covariances, whereas no such shrinkage is received by the whole-scan description. When the same shrinkage is applied to the whole-scan estimate, its AUROC is raised by 0.013 (unadjusted DeLong $p = 0.004$). The windowed description then differs from this matched baseline by -0.002 ($p = 0.709$), against 0.010 ($p = 0.281$) from the unshrunk baseline. The small advantage is, therefore, attributed to the regularisation rather than to variation within the scan. Comparisons between dynamic and static connectivity should accordingly match the regularisation of both descriptions.

Several secondary results bound what these data support. The ensemble probabilities are well calibrated, with an expected calibration error of 0.046 under pooled ten-fold validation and 0.040 under leave-one-site-out validation. Calibration, however, does not provide a useful guarantee on the answered cases. Specifically, the per-site abstention rule meets its target but answers only 170 of 871 participants (19.5%), and 143 of them are at a single site. Finally, the attainable accuracy against these labels lies between approximately 0.85 and 0.99. However, this ceiling cannot be fixed from two instruments applied to one population [27].

5.2. Limitations

The conclusions are limited by the composition of the cohort. Participants range from 6.5 to 58 years, with a median of 14.8 years, so the results do not bear on early identification. In addition, the primary cohort contains 727 male and 144 female participants. Moreover, the comparison group is typically developing rather than clinically referred, and every label is a diagnosis obtained before the scan. The results, therefore, describe discrimination within an aggregated research cohort rather than performance in a differential-diagnostic setting.

The evaluation is also limited to one dataset. Leave-one-site-out validation estimates transfer among the acquisition sites represented in ABIDE-I. It does not, however, estimate transfer to independent datasets, scanners, or preprocessing pipelines. For this reason, external validation on an independent cohort is still required before any clinical use is considered.

The group-difference analyses depend on the preprocessing and on their exploratory design. All results are obtained with one preprocessing strategy without global signal regression, one parcellation, and one nuisance model. The direction of the group differences is, therefore, conditional on these choices. In addition, the group differences are univariate contrasts computed without reference to the classifier. They describe how the groups differ, but not which measurements the classifier uses. Moreover, the motion-matched and regional analyses are exploratory, and the regional analysis is defined after inspection of the system-level results.

Several design choices further limit the comparisons. Only the logistic model is tuned, so the tree ensembles are represented by fixed configurations. In addition, the pairwise p-values are unadjusted and are interpreted descriptively. Several approaches, moreover, are evaluated but not adopted. Site harmonisation shows no evidence of benefit, within-scan variability lowers accuracy, and three atlases perform no better than

one. Similarly, demographic variables add about 0.004 AUROC, and restriction to the most discriminating regions lowers performance at every list length. These negative results are reported because each of them bounds what a subsequent study on this public dataset can expect.

5.3. Future work

Several extensions follow directly from these findings. The most direct extension is external validation on an independent cohort, such as ABIDE-II. Because the available ABIDE-II derivatives are produced by a different preprocessing pipeline, such a test requires either the same preprocessing or an explicit analysis of transfer across pipelines.

The dependence of detection on cognitive ability also requires further study. In particular, detection should be evaluated in samples that are balanced for cognitive ability. In addition, the dependence should be tested in cohorts with more cognitively able autistic participants, more female participants, and younger children. Clinically referred comparison groups would, moreover, move the evaluation closer to a differential-diagnostic setting.

The somatomotor difference should, in turn, be tested under alternative denoising strategies. These strategies include global signal regression and finer measures of head motion. In addition, tuned tree ensembles and other nonlinear models should be compared under the same nested protocol, so that the classifier comparison extends beyond fixed configurations.

Finally, the feature-selection result suggests a reporting standard for connectivity-based classification studies. Under this standard, every step fitted to data is placed inside the cross-validation loop and is declared together with the reported performance. Such reporting would allow published accuracies on ABIDE-I to be compared on an equal footing.

6. Conclusion

Connectivity-based classification of autism on ABIDE-I is examined with a single pipeline in which every learned quantity is fitted within the training partition. Under this protocol, a participant-pooled AUROC of 0.776 is obtained, and it is maintained when a whole site is withheld from training. In addition, accuracy at the level of published benchmarks is reached without any step fitted before the data are split.

Six findings are established on this basis. Detection is weakest in cognitively able autistic individuals, and somatomotor hypoconnectivity strengthens under motion matching. Moreover, connection-level feature selection before the split inflates reported performance, and linear classifiers outperform fixed-configuration tree ensembles. In addition, classifier scores are not associated with the degree of autism within the autistic group, and the advantage of windowed connectivity is explained by the covariance shrinkage that this description applies. Together, these findings indicate that reported performance on ABIDE-I depends strongly on sample composition and analysis procedure. They also identify a somatomotor connectivity difference in autism that is robust to head motion.

References

[1] Abraham, A., Milham, M.P., Di Martino, A., Craddock, R.C., Samaras, D., Thirion, B., Varoquaux, G., 2017. Deriving reproducible biomarkers from multi-site resting-state data: An autism-based example. NeuroImage 147, 736–745. URL: https://doi.org/10.1016/j.neuroimage.2016.10.045, doi:10.1016/j.neuroimage.2016.10.045.

[2] Allen, E.A., Damaraju, E., Plis, S.M., Erhardt, E.B., Eichele, T., Calhoun, V.D., 2014. Tracking whole-brain connectivity dynamics in the resting state. Cerebral Cortex 24, 663–676.

[3] Almuqhim, F., Saeed, F., 2021. Asd-saenet: A sparse autoencoder, and deep-neural network model for detecting autism spectrum disorder (asd) using fmri data. Front. Comput. Neurosci. 15, 654315. URL: https://doi.org/10.3389/fncom.2021.654315, doi:10.3389/fncom.2021.654315.

[4] Angelopoulos, A.N., Bates, S., 2023. Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning 16, 494–591.

[5] Angelopoulos, A.N., Bates, S., Candès, E.J., Jordan, M.I., Lei, L., 2021. Learn then test: Calibrating predictive algorithms to achieve risk control. arXiv preprint arXiv:2110.01052 .

[6] Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B 57, 289–300.

[7] Breiman, L., 2001. Random forests. Machine Learning 45, 5–32.

[8] Chen, C.P., Keown, C.L., Jahedi, A., Nair, A., Pflieger, M.E., Bailey, B.A., Müller, R.A., 2015. Diagnostic classification of intrinsic functional connectivity highlights somatosensory, default mode, and visual regions in autism. NeuroImage: Clinical 8, 238–245.

[9] Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794.

[10] Clopper, C.J., Pearson, E.S., 1934. The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika 26, 404–413. URL: https://doi.org/10.1093/biomet/26.4.404, doi:10.1093/biomet/26.4.404.

[11] Constantino, J.N., Gruber, C.P., 2005. Social Responsiveness Scale. Western Psychological Services.

[12] Craddock, C., Benhajali, Y., Chu, C., Chouinard, F., Evans, A., Jakab, A., et al., 2013. The neuro bureau preprocessing initiative: Open sharing of preprocessed neuroimaging data and derivatives. Front. Neuroinform. 7.

[13] Craddock, R.C., James, G.A., Holtzheimer, P.E., Hu, X.P., Mayberg, H.S., 2012. A whole brain fMRI atlas generated via spatially constrained spectral clustering. Human Brain Mapping 33, 1914–1928.

[14] Dadi, K., Rahim, M., Abraham, A., Chyzhyk, D., Milham, M., Thirion, B., Varoquaux, G., 2019. Benchmarking functional connectome-based predictive models for resting-state fmri. NeuroImage 192, 115–134. URL: https://doi.org/10.1016/j.neuroimage.2019.02.062, doi:10.1016/j.neuroimage.2019.02.062.

[15] DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L., 1988. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. Biometrics 44, 837–845.

[16] Demšar, J., 2006. Statistical comparisons of classifiers over multiple data sets. J. Mach. Learn. Res. 7, 1–30. URL: https://www.jmlr.org/papers/v7/demsar06a.html.

[17] DerSimonian, R., Laird, N., 1986. Meta-analysis in clinical trials. Controlled Clinical Trials 7, 177–188.

[18] Desikan, R.S., Ségonne, F., Fischl, B., Quinn, B.T., Dickerson, B.C., Blacker, C.J., et al., 2006. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. NeuroImage 31, 968–980.

[19] Di Martino, A., Yan, C.G., Li, Q., Denio, E., Castellanos, F.X., Alaerts, K., Anderson, J.S., Assaf, M., Bookheimer, S.Y., Dapretto, M., Deen, B., Delmonte, S., Dinstein, I., Ertl-Wagner, B., Fair, D.A., Gallagher, L., Kennedy, D.P., Keown, C.L., Keysers, C., Milham, M.P., 2014. The autism brain imaging data exchange: Towards a large-scale evaluation of the intrinsic brain architecture in autism. Mol. Psychiatry 19, 659–667. URL: https://doi.org/10.1038/mp.2013.78, doi:10.1038/mp.2013.78.

[20] Eslami, T., Mirjalili, V., Fong, A., Laird, A.R., Saeed, F., 2019. Asd-diagnet: A hybrid learning approach for detection of autism spectrum disorder using fmri data. Front. Neuroinform. 13, 70. URL: https://doi.org/10.3389/fninf.2019.00070, doi:10.3389/fninf.2019.00070.

[21] Fortin, J.P., Cullen, N., Sheline, Y.I., Taylor, W.D., Aselcioglu, I., Cook, P.A., Adams, P., Cooper, C., Fava, M., McGrath, P.J., McInnis, M., Phillips, M.L., Trivedi, M.H., Weissman, M.M., Shinohara, R.T., 2018. Harmonization of cortical thickness measurements across scanners and sites. NeuroImage 167, 104–120. URL: https://doi.org/10.1016/j.neuroimage.2017.11.024, doi:10.1016/j.neuroimage.2017.11.024.

[22] Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks, in: Adv. Neural Inf. Process. Syst., pp. 4878–4887. URL: https://papers.nips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html.

[23] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proc. 34th Int. Conf. Mach. Learn., pp. 1321–1330. URL: https://proceedings.mlr.press/v70/guo17a.html.

[24] Heinsfeld, A.S., Franco, A.R., Craddock, R.C., Buchweitz, A., Meneguzzi, F., 2018. Identification of autism spectrum disorder using deep learning and the abide dataset. NeuroImage Clin. 17, 16–23. URL: https://doi.org/10.1016/j.nicl.2017.08.017, doi:10.1016/j.nicl.2017.08.017.

[25] Holiga, Š., Hipp, J.F., Chatham, C.H., Garces, P., Spooren, W., D'Ardhuy, X.L., Bertolino, A., Bouquet, C., Buitelaar, J.K., Bours, C., et al., 2019. Patients with autism spectrum disorders display reproducible functional connectivity alterations. Science Translational Medicine 11, eaat9223.

[26] Huda, S., Khan, D.M., Masroor, K., Warda, Rashid, A., Shabbir, M., 2024. Advancements in automated diagnosis of autism spectrum disorder through deep learning and resting-state functional mri biomarkers: a systematic review. Cogn. Neurodyn. 18, 3585–3601. URL: https://doi.org/10.1007/s11571-024-10176-z, doi:10.1007/s11571-024-10176-z.

[27] Hui, S.L., Walter, S.D., 1980. Estimating the error rates of diagnostic tests. Biometrics 36, 167–171.

[28] Hutchison, R.M., Womelsdorf, T., Allen, E.A., Bandettini, P.A., Calhoun, V.D., Corbetta, M., Della Penna, S., Duyn, J.H., Glover, G.H., Gonzalez-Castillo, J., Handwerker, D.A., Keilholz, S., Kiviniemi, V., Leopold, D.A., de Pasquale, F., Sporns, O., Walter, M., Chang, C., 2013. Dynamic functional connectivity: Promise, issues, and interpretations. NeuroImage 80, 360–378. URL: https://doi.org/10.1016/j.neuroimage.2013.05.079, doi:10.1016/j.neuroimage.2013.05.079.

[29] Ilioska, I., Oldehinkel, M., Llera, A., Chopra, S., Looden, T., Chauvin, R., Van Rooij, D., Floris, D.L., Tillmann, J., Moessnang, C., et al., 2023. Connectome-wide mega-analysis reveals robust patterns of atypical functional connectivity in autism. Biological Psychiatry 94, 29–39.

[30] Jenkinson, M., Bannister, P., Brady, M., Smith, S., 2002. Improved optimization for the robust and accurate linear registration and motion correction of brain images. NeuroImage 17, 825–841.

[31] Johnson, W.E., Li, C., Rabinovic, A., 2007. Adjusting batch effects in microarray expression data using empirical bayes methods. Biostatistics 8, 118–127. URL: https://doi.org/10.1093/biostatistics/kxj037, doi:10.1093/biostatistics/kxj037.

[32] Kawahara, J., Brown, C.J., Miller, S.P., Booth, B.G., Chau, V., Grunau, R.E., et al., 2017. Brainnetcnn: Convolutional neural networks for brain networks; towards predicting neurodevelopment. NeuroImage 146, 1038–1049.

[33] Ledoit, O., Wolf, M., 2004. A well-conditioned estimator for large-dimensional covariance matrices. J. Multivar. Anal. 88, 365–411. URL: https://doi.org/10.1016/S0047-259X(03)00096-4, doi:10.1016/S0047-259X(03)00096-4.

[34] Leonardi, N., Van De Ville, D., 2015. On spurious and real fluctuations of dynamic functional connectivity during rest. NeuroImage 104, 430–436. URL: https://doi.org/10.1016/j.neuroimage.2014.09.007, doi:10.1016/j.neuroimage.2014.09.007.

[35] Liu, X., Hasan, M.R., Gedeon, T., Hossain, M.Z., 2024. Made-for-asd: A multi-atlas deep ensemble network for diagnosing autism spectrum disorder. Comput. Biol. Med. 182, 109083. URL: https://doi.org/10.1016/j.compbiomed.2024.109083, doi:10.1016/j.compbiomed.2024.109083.

[36] Lord, C., Risi, S., Lambrecht, L., Cook, E.H., Leventhal, B.L., DiLavore, P.C., et al., 2000. The autism diagnostic observation schedule–generic: A standard measure of social and communication deficits associated with the spectrum of autism. J. Autism Dev. Disord. 30, 205–223.

[37] Lord, C., Rutter, M., Le Couteur, A., 1994. Autism diagnostic interview–revised: A revised version of a diagnostic interview for caregivers of individuals with possible pervasive developmental disorders. Journal of Autism and Developmental Disorders 24, 659–685.

[38] Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Adv. Neural Inf. Process. Syst., pp. 4765–4774. URL: https://papers.nips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html.

[39] Maenner, M.J., Warren, Z., Williams, A.R., Amoakohene, E., Bakian, A.V., Bilder, D.A., et al., 2023. Prevalence and characteristics of autism spectrum disorder among children aged 8 years—autism and developmental disabilities monitoring network, 11 sites, united states, 2020. MMWR Surveill. Summ. 72, 1–14.

[40] McNemar, Q., 1947. Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika 12, 153–157.

[41] Menon, A., van Rooyen, B., Ong, C.S., Williamson, R.C., 2015. Learning from corrupted binary labels via class-probability estimation, in: Proceedings of the 32nd International Conference on Machine Learning, pp. 125–134.

[42] Naeini, M.P., Cooper, G.F., Hauskrecht, M., 2015. Obtaining well calibrated probabilities using Bayesian binning. Proceedings of the AAAI Conference on Artificial Intelligence 29, 2901–2907.

[43] Nielsen, J.A., Zielinski, B.A., Fletcher, P.T., Alexander, A.L., Lange, N., Bigler, E.D., Lainhart, J.E., Anderson, J.S., 2013. Multisite functional connectivity MRI classification of autism: ABIDE results. Frontiers in Human Neuroscience 7, 599.

[44] Parisot, S., Ktena, S.I., Ferrante, E., Lee, M., Guerrero, R., Glocker, B., Rueckert, D., 2018. Disease prediction using graph convolutional networks: Application to autism spectrum disorder and alzheimer's disease. Med. Image Anal. 48, 117–130. URL: https://doi.org/10.1016/j.media.2018.06.001, doi:10.1016/j.media.2018.06.001.

[45] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

[46] Power, J.D., Barnes, K.A., Snyder, A.Z., Schlaggar, B.L., Petersen, S.E., 2012. Spurious but systematic correlations in functional connectivity MRI networks arise from subject motion. NeuroImage 59, 2142–2154.

[47] Preti, M.G., Bolton, T.A.W., Van De Ville, D., 2017. The dynamic functional connectome: State-of-the-art and perspectives. NeuroImage 160, 41–54. URL: https://doi.org/10.1016/j.neuroimage.2016.12.061, doi:10.1016/j.neuroimage.2016.12.061.

[48] Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V., Gulin, A., 2018. CatBoost: Unbiased boosting with categorical features, in: Advances in Neural Information Processing Systems, pp. 6638–6648.

[49] Santana, C.P., de Carvalho, E.A., Rodrigues, I.D., Bastos, G.S., de Souza, A.D., de Brito, L.L., 2022. rs-fmri and machine learning for asd diagnosis: a systematic review and meta-analysis. Sci. Rep. 12, 6030. URL: https://doi.org/10.1038/s41598-022-09821-6, doi:10.1038/s41598-022-09821-6.

[50] Tang, J., Chen, J., Hu, M., Hu, Y., Zhang, Z., Xiao, L., 2025. Diagnosis of autism spectrum disorder (asd) by dynamic functional connectivity using gnn-lstm. Sensors 25, 156. URL: https://doi.org/10.3390/s25010156, doi:10.3390/s25010156.

[51] Tang, Y., Yang, C., Xu, Y., Zhang, H., Xie, H., 2026. Cross-attention-guided subject-adaptive graph learning for multimodal autism classification: integrating structural and functional mri data. Brain Inform. 13, 32. URL: https://doi.org/10.1186/s40708-026-00318-w, doi:10.1186/s40708-026-00318-w.

[52] Tzourio-Mazoyer, N., Landeau, B., Papathanassiou, D., Crivello, F., Etard, O., Delcroix, N., et al., 2002. Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain. NeuroImage 15, 273–289.

[53] Van Calster, B., Wynants, L., Verbeek, J.F.M., Verbakel, J.Y., Christodoulou, E., Vickers, A.J., Roobol, M.J., Steyerberg, E.W., 2018. Reporting and interpreting decision curve analysis: A guide for investigators. Eur. Urol. 74, 796–804. URL: https://doi.org/10.1016/j.eururo.2018.08.038, doi:10.1016/j.eururo.2018.08.038.

[54] Varoquaux, G., Baronnet, F., Kleinschmidt, A., Fillard, P., Thirion, B., 2010. Detection of brain functional-connectivity difference in post-stroke patients using group-level covariance modeling, in: Medical Image Computing and Computer-Assisted Intervention – MICCAI 2010, pp. 200–208. URL: https://doi.org/10.1007/978-3-642-15705-9_25, doi:10.1007/978-3-642-15705-9_25.

[55] Vidya, S., Gupta, K., Aly, A., Wills, A., Ifeachor, E., Shankar, R., 2025. Identification of critical brain regions for autism diagnosis from fmri data using explainable ai: an observational analysis of the abide dataset. eClinicalMedicine 88, 103452. URL: https://doi.org/10.1016/j.eclinm.2025.103452, doi:10.1016/j.eclinm.2025.103452.

[56] Wilson, E.B., 1927. Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association 22, 209–212.

[57] Yeo, B.T.T., Krienen, F.M., Sepulcre, J., Sabuncu, M.R., Lashkari, D., Hollinshead, M., Roffman, J.L., Smoller, J.W., Zollei, L., Polimeni, J.R., Fischl, B., Liu, H., Buckner, R.L., 2011. The organization of the human cerebral cortex estimated by intrinsic functional connectivity. J. Neurophysiol. 106, 1125–1165. URL: https://doi.org/10.1152/jn.00338.2011, doi:10.1152/jn.00338.2011.

[58] Zhou, H., Li, L., 2014. Regularized matrix regression. J. R. Stat. Soc. Ser. B Stat. Methodol. 76, 463–483. URL: https://doi.org/10.1111/rssb.12031, doi:10.1111/rssb.12031.